

**SỞ GIÁO DỤC VÀ ĐÀO TẠO NAM ĐỊNH  
TRƯỜNG THPT XUÂN TRƯỜNG B**

**BÁO CÁO KINH NGHIỆM**  
**GIẢNG DẠY TOÁN BẰNG TIẾNG ANH**  
**CHƯƠNG "Dãy số - Cấp số cộng - Cấp số nhân"**

**EXPERIENCE REPORT**  
**FOR TEACHING MATHEMATICS IN ENGLISH OF CHAPTER**  
**"SEQUENCES - ARITHMETIC SEQUENCES - GEOMETRIC SEQUENCES"**

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**MỘT SỐ KINH NGHIỆM GIẢNG DẠY TOÁN BẰNG TIẾNG ANH  
CHƯƠNG "DÃY SỐ - CẤP SỐ CỘNG - CẤP SỐ NHÂN"**

## 2. Lĩnh vực áp dụng

Sáng kiến được áp dụng trong giảng dạy nội dung "Dãy số - Cấp số cộng - Cấp số nhân" (chương III – Giải tích 11) cho hai đối tượng: Một là học sinh lớp 11 (trong giảng dạy môn Toán bằng tiếng Anh); Hai là sử dụng cho giáo viên vừa làm tư liệu, vừa định hướng sáng tạo khi dạy học nội dung dãy số - cấp số cộng - cấp số nhân bằng tiếng Anh.

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# Chương 1

## Điều kiện, hoàn cảnh tạo ra sáng kiến

Hình thành và rèn luyện năng lực học tập bộ môn, liên môn là một yêu cầu tất yếu của mỗi môn học ở cấp học phổ thông. Trong quá trình giảng dạy cho nhiều đối tượng học sinh các năm học từ 2013 đến nay, chúng tôi thấy cần thiết phải phát triển cho học sinh năng lực Toán học để giúp học sinh nắm bắt và làm chủ được các phương pháp và kĩ thuật giải toán đa dạng. Điều này giúp học sinh tích cực hơn trong việc học của mình, gợi động cơ yêu thích môn học và đáp ứng được các mức độ yêu cầu khác nhau của các kì thi.

Bên cạnh đó, với sự phát triển của mạng internet và các khóa học trực tuyến (MOOC), khoảng cách địa lý không còn là vấn đề khó khăn, chúng ta có thể ngồi tại nhà ở Việt Nam và theo dõi các bài giảng của các giảng viên ở khắp nơi trên thế giới. Tuy nhiên, ngoại ngữ luôn là một trở ngại lớn đối với chúng ta nói chung và học sinh nói riêng, đặc biệt là khả năng giao tiếp.

Dạy học môn Toán và các môn khoa học tự nhiên bằng tiếng Anh là một trong những nhiệm vụ trọng tâm của đề án 1400 và 959 đã được Thủ tướng Chính phủ phê duyệt.

Thực hiện công văn số: 1304/SGDĐT-GDTrH của Sở Giáo dục và Đào tạo Nam Định về việc: Hướng dẫn giảng dạy, tổ chức Hội giảng và Hội thi Toán và các môn KHTN bằng tiếng Anh năm học 2015-2016; căn cứ vào kế hoạch và tình hình thực tế của trường THPT Xuân Trường B; chúng tôi viết sáng kiến: **Một số kinh nghiệm trong giảng dạy Toán bằng tiếng Anh chương "dãy số – cấp số cộng – cấp số nhân"** với ba lí do chính sau:

- **Xu thế hội nhập:** Theo Nghị quyết 29 ngày 4 tháng 11 năm 2013 của Ban Chấp Hành Trung ương Đảng: "Chủ động hội nhập quốc tế về giáo dục, đào tạo trên cơ sở giữ vững độc lập, tự chủ, bảo đảm định hướng xã hội chủ nghĩa, bảo tồn và phát huy các giá trị văn hóa tốt đẹp của dân tộc, tiếp thu có chọn lọc tinh hoa văn hóa và thành tựu khoa học, công nghệ của nhân loại. Hoàn thiện cơ chế hợp tác song phương và đa phương, thực hiện các cam kết quốc tế về giáo dục, đào tạo". Đặc biệt, trong môn Toán, mảng kiến thức về dãy số, cấp số cộng, cấp số nhân đều xuất hiện trong chương trình của Ấn Độ, Canada, Mỹ... , và đây cũng là mảng kiến thức bắt buộc trong các kì thi SAT, ACT.
- **Kiến thức Toán:** Kiến thức chương này đơn giản, dễ hiểu; không liên quan nhiều đến các mảng kiến thức khác của chương trình phổ thông; và đặc biệt, có nhiều ví dụ, bài toán

thực tế; tuy nhiên vẫn đảm bảo phát triển những năng lực tư duy quan trọng mà chương trình giáo dục phổ thông đặt ra.

- **Tính liên môn, tích hợp:** Chương dãy số có rất nhiều bài toán liên môn, các bài toán ứng dụng trong thực tiễn cuộc sống, trong các bộ môn Hóa, Lý... đặc biệt là phần phương pháp quy nạp xuất hiện nhiều trong môn Tin (liên quan đến các giải thuật đệ quy: các bài toán sắp xếp...); và tất nhiên, phải kể đến tính liên môn với môn Tiếng Anh, giúp học sinh bước đầu sử dụng tiếng Anh trong cuộc sống, tạo hứng thú trong việc học ngoại ngữ.

Sáng kiến này tập trung vào trình bày một cách có hệ thống kiến thức và bài tập của chương dãy số trong chương trình Toán lớp 11 với mục đích:

- Thực hiện nguyên tắc học ngoại ngữ “Learning English through usage” (học tiếng Anh qua sử dụng tiếng). Cách học này nhằm khắc phục nhược điểm “học nhưng không sử dụng được” của một bộ phận lớn học sinh; thông qua đó cũng hỗ trợ cho chính việc học tiếng Anh của học sinh.
- Tóm tắt lí thuyết các nội dung: Phương pháp quy nạp, dãy số, cấp số cộng, cấp số nhân.
- Lời giải mẫu chi tiết các ví dụ tương ứng với từng nội dung; có những ví dụ mở rộng kiến thức so với SGK, phục vụ cho các kì thi tuyển sinh, đặc biệt là thi học sinh giỏi.
- Cung cấp hệ thống bài tập phong phú, đặc biệt là các bài tập từ những cuộc thi trên thế giới, các bài tập mang tính thực tiễn; hầu hết bài tập đều có gợi ý, hướng dẫn.

Hi vọng tài liệu sẽ là một kênh tham khảo cho các em học sinh và quý đồng nghiệp, giúp giảm bớt công sức thời gian trong việc tìm tòi, sưu tầm tài liệu về dãy số.

Vì trình độ ngoại ngữ còn hạn chế nên không thể tránh khỏi sai sót, chúng tôi rất mong nhận được ý kiến đóng góp của quý thầy cô và các em học sinh.

## Chương 2

# Thực trạng trước khi tạo ra sáng kiến

Trong quá trình soạn bài, giảng dạy Toán bằng tiếng Anh, chúng tôi thấy nổi bật nên những khó khăn sau:

- Đối với học sinh: Các em lúng túng khi đọc các công thức toán; không tự tin khi giao tiếp; trình bày lời giải chưa chuẩn: sai ngữ pháp, dùng từ chuyên ngành không chuẩn...
- Đối với giáo viên: Khó khăn trong tra cứu các thuật ngữ chuyên ngành, khó tìm kiếm bài tập, không có lời giải mẫu cho các dạng toán, phát âm chưa chuẩn...
- Bài tập chưa phong phú, không có tính hệ thống; chưa có cách trình bày, lời giải mẫu... cho từng dạng toán.

Ngoài ra, trên mạng internet có rất nhiều bài giảng của các trường Đại học, Trung học... ở khắp nơi trên thế giới bằng tiếng Anh; tuy nhiên giáo viên và học sinh của ta chưa khai thác hết được nguồn tài nguyên khổng lồ này. Lí do chính, là vấn đề ngoại ngữ!

Vì vậy, chúng tôi viết sáng kiến này để phần nào giải quyết những khó khăn trên.

Lựa chọn mảng kiến thức nào để giảng dạy bằng tiếng Anh luôn là một câu hỏi được chúng tôi đặt ra đầu tiên. Bài tập về phương pháp quy nạp, dãy số, cấp số là nội dung xuất hiện trong các đề thi giữa học kỳ II, thi cuối năm, thi tuyển sinh và đặc biệt nó cũng xuất hiện trong các đề thi HSG. Loại bài tập này không quá phong phú về phương pháp và kĩ thuật, nhưng vẫn thể hiện mối liên hệ mật thiết giữa nhiều mảng kiến thức Toán (mệnh đề tập hợp, phương trình, hệ phương trình), do đó nó là một trong những nội dung giúp rèn luyện và phát triển một số năng lực Toán học cho người dạy và người học:

- Năng lực tư duy logic, sử dụng ngôn ngữ (tiếng Việt, tiếng Anh) và kí hiệu chính xác.
- Năng lực suy đoán và tưởng tượng, liên hệ.
- Năng lực làm việc theo quy trình.
- Những hoạt động trí tuệ cơ bản: Phân tích, tổng hợp, khái quát hóa. . .

- Hình thành những phẩm chất trí tuệ có ích trong học tập, trong công tác và cuộc sống: Tính linh hoạt, khả năng lật ngược vấn đề, tự phản biện, tính độc lập, tính sáng tạo.

Hơn nữa, nội dung kiến thức chương này đảm bảo *ba* yếu tố, mà theo chúng tôi là quan trọng:

- Kiến thức đơn giản, dễ hiểu; từ vựng không nhiều.
- Mảng kiến thức chương này đều xuất hiện trong chương trình phổ thông của các nước: Canada, Mỹ, Ấn Độ, Nam Phi..., trong các kì thi SAT, ACT.
- Có nhiều bài toán thực tế, nhiều bài toán liên môn (Hóa, Lý, Tin, Sinh)

Trước khi giảng dạy chương "Dãy số, cấp số cộng, cấp số nhân", chúng tôi đã tìm hiểu nội dung tương ứng trong các SGK của một số nước và trên các diễn đàn, các website của nước ngoài. Sau đó, tôi xây dựng bài dạy dựa trên chương trình SGK của Việt Nam, cụ thể:

- Dạy nội dung tương ứng bằng tiếng Việt.
- Lập danh sách các từ khóa, thuật ngữ của chương, tra các thuật ngữ tương ứng trong tiếng Anh; và dựa vào các từ khóa này tìm bài tập trên mạng internet.
- Cung cấp danh sách các từ khóa này cho học sinh, nhờ các giáo viên tiếng Anh hướng dẫn các em tập phát âm.
- Cung cấp hệ thống đề bài tập bằng tiếng Anh, giao cho các học sinh về nhà dịch và làm bằng tiếng Việt.
- Chia nhóm để các em thảo luận cách trình bày bằng tiếng Anh tại lớp.
- Dạy một bài hoàn toàn bằng tiếng Anh.

# Chương 3

## Giải pháp

### 3.1 Tóm tắt nội dung giải pháp

Nội dung chương "Dãy số - Cấp số cộng - Cấp số nhân" trong chương trình SGK 11 gồm những nội dung:

1. Phương pháp quy nạp toán học.
2. Dãy số.
3. Cấp số cộng.
4. Cấp số nhân.

Trong chương này, chúng tôi trình bày hai nội dung:

- Cung cấp hệ thống mẫu câu, cấu trúc câu, từ vựng thường sử dụng của môn Toán và đặc biệt là của chương dãy số.
- Hệ thống lại kiến thức về phương pháp quy nạp, dãy số, cấp số; cung cấp các bài toán cơ bản với lời giải mẫu chi tiết; hệ thống bài tập vận dụng phong phú, đầy đủ có gợi ý; giới thiệu các bài tập khó từ những cuộc thi trên thế giới.

Mỗi một bài gồm có tóm tắt lí thuyết, các ví dụ được trình bày lời giải chi tiết và hệ thống bài tập vận dụng. Trong đó, có những điểm nổi bật sau:

- Phần phương pháp quy nạp, tôi giới thiệu cả các ví dụ về phương pháp quy nạp tổng quát, bên cạnh đó tôi giới thiệu một số ví dụ, bài tập kinh điển của phương pháp quy nạp và những bài toán vận dụng.
- Phần dãy số, có một số bài về dãy Fibonacci, về dãy cho bằng hệ thức truy hồi phụ thuộc vào từ hai số hạng đứng trước trở lên...
- Phần cấp số có nhiều bài tập thực tế, một số bài tập liên môn và các bài từ những cuộc thi trên thế giới.

Phần này xin được trình bày bằng tiếng Anh, các bài tập do tôi lấy từ hai nguồn chính: tự dịch từ các bài toán tiếng Việt và tài liệu trên mạng internet (các đề thi trên thế giới).

## 3.2 Nội dung của giải pháp

### 3.2.1 Một số mẫu câu tiếng Anh thường dùng

- It follows from... that...: Từ... suy ra...
- We deduce from... that...: Ta suy ra từ... rằng...
- Conversely,... implies that...: Ngược lại,... có nghĩa...
- Equality (1) holds, by Proposition 2: Theo mệnh đề 2, đẳng thức (1) đúng.
- By definition,...: Theo định nghĩa...
- The following statements are equivalent: Những phát biểu sau là tương đương.
- Thanks to... the properties... and... of... are equivalent to each other: Nhờ... những tính chất... là tương đương.
- ... has the following properties:... có những tính chất sau.
- Theorem 1 holds unconditionally: Định lý 1 được suy ra một cách hiển nhiên
- This result is conditional on Axiom A: Kết quả này được suy ra từ tiên đề A...
- ... is an immediate consequence of Theorem 3: ... là hệ quả trực tiếp từ định lý 3.
- Note that... is well-defined, since...: Chú ý rằng... luôn đúng vì...
- As... satisfies... formula (1) can be simplified as follows: Vì... thỏa mãn... công thức (1) có thể được viết đơn giản như sau.
- We conclude (the argument) by combining in equalities (1) and (2): Từ (1) và (2) ta suy ra điều phải chứng minh.
- (Let us) denote by  $X$  the set of all...: Ký hiệu  $X$  là tập hợp...
- Let  $X$  be the set of all...: Lấy  $X$  là tập hợp tất cả các...
- Recall that... by assumption: Theo giả thiết ta có...
- It is enough to show that...: Điều kiện đủ là...
- We are reduced to proving that...: Suy ra ta cần chứng minh rằng...
- The main idea is as follows...: Ý tưởng chính là như sau...
- We argue by contradiction/Assume that ... exists: Giả sử phản chứng là...

- The formal argument proceeds in several steps: Kết luận được đưa ra từ các bước sau. . .
- Consider first the special case when...: Xét trường hợp đặc biệt đầu tiên. . .
- The assumptions ... and ... are independent (of each other) since...: Các giả sử... và... là độc lập nhau vì. . .
- ... which proves the required claim: ... điều cần chứng minh.
- We use induction on  $n$  to show that...: Ta sử dụng phương pháp chứng minh quy nạp với  $n$  để chỉ ra rằng...
- On the other hand,...: Một mặt,...
- ... which mean that...: điều đó chứng tỏ rằng. . .
- In others word,... nói một cách khác...

### 3.2.2 Từ vựng Toán bằng tiếng Anh thường gặp

- |                                                             |                                                |
|-------------------------------------------------------------|------------------------------------------------|
| • argument (n): lập luận                                    | • consist (v): gồm có                          |
| • assume (suppose) (v): giả sử                              | • contradict: mâu thuẫn với, trái với          |
| • assumption (n): sự giả sử                                 | • contradiction (n): sự phủ định, sự mâu thuẫn |
| • axiom (n): tiên đề                                        | • conversely (adv): ngược lại                  |
| • case (n): khả năng, trường hợp                            | • corollary (n): hệ quả                        |
| • special case: cách đặc biệt                               | • deduce (v): suy ra                           |
| • claim (n): đòi hỏi, yêu cầu                               | • derive (v): suy ra                           |
| • concept (n): khái niệm                                    | • distinct (adj): riêng biệt, phân biệt        |
| • conclude (v): kết luận                                    | • domain (n): miền xác định                    |
| • conclusion (n): sự kết luận                               | • element (n): phần tử                         |
| • a necessary and sufficient condition: điều kiện cần và đủ | • equation (n): phương trình                   |
| • conjecture (n): sự giả định, giả sử                       | • equivalent (adj): tương đương                |
| • consequence (n): hệ quả, kết quả                          | • establish (v): thiết lập                     |
| • consider (v): xét, chú ý đến cho rằng                     | • explain (v): giải thích                      |
|                                                             | • expression (n): biểu thức                    |

- false (adj): sai
- form (v): hình thành, tạo thành
- hold (v): xảy ra
- hence (adv): sau đây, kể từ đây
- if and only if (iff): khi và chỉ khi
- inequality (n): bất đẳng thức
- imply: kéo theo, suy ra
- induction (n): phép quy nạp
- internal (adj): ở trong, nội bộ
- lemma (n): bổ đề
- nested (adj): được lồng nhau
- observe (v): quan sát, nhận xét
- obtain (v): nhận được
- obviously (adv): một cách rõ ràng
- on one hand: một mặt
- on the other hand: mặt khác
- proof (n): bằng chứng
- satisfy property: thỏa mãn tính chất
- proposition (n): mệnh đề
- reasoning (n): sự biện luận
- reduce (v): quy về, rút gọn
- side (n): cạnh, vế (trái, phải)
- remark (n): chú ý, chú thích
- set (v): đặt
- set (n): tập hợp
- subset (n): tập hợp con
- substitute (v): thay thế
- such that: sao cho
- statement (n): mệnh đề
- similarly (adv): tương tự
- equivalent to (adj): tương đương với
- theorem (n): định lí
- therefore (adv): bởi vậy, cho nên
- true (adj): đúng
- thus (adv): như vậy, như thế
- truth (n): chân lý
- vein (n): lối, cách
- verify (v): kiểm tra lại, thử lại
- wlog (without loss of generality): không mất tính tổng quát
- yield (v): thu được, cho

### 3.2.3 Từ vựng Toán chương "Dãy số"

Đầu tiên, chúng tôi dạy học sinh các kiến thức liên quan đến dãy số, cấp số cộng, cấp số nhân bằng tiếng Việt. Tiếp theo, chúng tôi cung cấp cho các em các từ khóa liên quan đến dãy số bằng tiếng Anh. Phần này, có thể nhờ các thầy cô tiếng Anh hướng dẫn các em phát âm.

- add (v): cộng, thêm
- bounded (adj): bị chặn
  - ✓ — above: bị chặn trên
  - ✓ — below: bị chặn dưới
- common difference (n): công sai
- common ratio (n): công bội
- conjecture (n): sự phỏng đoán
- consecutive (adj): liên tiếp
- define (v): định nghĩa
- decreasing (adj): giảm, nghịch biến
- determine (v): xác định
- divisible (adj): chia hết
- finite (adj): hữu hạn
- formula (n): công thức
- hypothesis (n): giả thuyết
- infinite (adj): vô hạn
- induction (n): quy nạp
- increasing (adj): tăng, đồng biến
- integer (n): số nguyên
- monotone (adj): đơn điệu
- multiple (n): bội số
- natural (adj): tự nhiên
- negative (adj): (số) âm
- order (n): thứ tự, trật tự
- positive (adj): (số) dương
- product (n): tích
- prime (adj): số nguyên tố
- progression (n): dãy số, cấp số
- recursive (adj): truy hồi, đệ quy
- sequence (n): dãy số
  - ✓ arithmetic — cấp số cộng
  - ✓ geometric — cấp số nhân
- step (n): bước
- subtract (v): trừ
- sum (n): tổng
- term (n): số hạng
  - ✓ first — số hạng đầu tiên
  - ✓ last — số hạng cuối cùng
  - ✓ general — số hạng tổng quát

### TÓM TẮT KIẾN THỨC – HỆ THỐNG BÀI TẬP CHƯƠNG "DÃY SỐ – CẤP SỐ CỘNG – CẤP SỐ NHÂN"

#### 3.2.4 Principle of Mathematical Induction

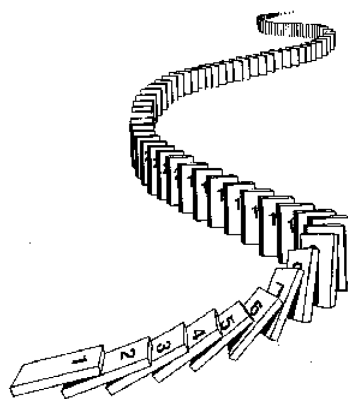
The Principle of Mathematical Induction is based on the following fairly intuitive observation. Suppose that we are to perform a task that involves a certain finite number of steps. Suppose that these steps are sequential. Finally, suppose that we know how to perform the  $n^{th}$  step provided we have accomplished the  $(n - 1)^{th}$  step. Thus if we are ever able to start the task (that is, if we

have a base case), then we should be able to finish it (because starting with the base we go to the next case, and then to the case following that, etc.). We formulate the Principle of Mathematical Induction as follows: **Principle of Mathematical Induction** Suppose we have an assertion  $P(n)$  concerning natural numbers satisfying the following two properties:

- **Base case:**  $P(k_0)$  is true for some natural number  $k_0$ ,
- **Induction step:** If  $P(k-1)$  is true then  $P(k)$  is true.

By the principle of induction, the assertion  $P(n)$  is true for every  $n \geq k_0$ .

One can liken the principle of mathematical induction to the domino effect. We imagine an infinite set of dominoes all lined up.



Consider an infinite sequence of dominoes, labeled  $1, 2, 3, \dots$ , where each domino is standing. Let  $P(n)$  is the proposition that the  $n^{\text{th}}$  domino is knocked over. We know that the first domino is knocked down, i.e.,  $P(1)$  is true. We also know that if whenever the  $k^{\text{th}}$  domino is knocked over, it knocks over the  $(k+1)^{\text{st}}$  domino, i.e.,  $P(k) \Rightarrow P(k+1)$  is true for all positive integers  $k$ . Hence, all dominoes are knocked over.  $P(n)$  is true for all positive integers  $n$ .

Sometimes, we need "Strong Induction", which can be useful in situations where assuming  $P(k)$  is true does not neatly lend itself to forcing  $P(k+1)$  to be true:

- **Base case:** Prove that  $P(k_0)$  and  $P(k_0+1)$  are true for some natural number  $k_0$ ,
- **Induction step:** Suppose that  $P(k)$  and  $P(k+1)$  are true for some natural number  $k \geq k_0$ , we prove that  $P(k+2)$  is true.

or

- **Base case:** Prove that  $P(k_0)$  is true for some natural number  $k_0$ ,
- **Induction step:** Suppose that  $P(k_0), P(k_0+1), \dots, P(k)$  are true for some natural number  $k \geq k_0$ , we prove that  $P(k+1)$  is true.

By the principle of induction, the assertion  $P(n)$  is true for every  $n \geq k_0$ .

**EXAMPLE****Example 1.** Prove that

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2} \quad (3.1)$$

for any integer  $n \geq 1$ .*Proof.* For  $n = 1$ , (3.1) is true, since  $1 = \frac{1(1+1)}{2}$ .Suppose that (3.1) is true for  $n = k \geq 1$ , that is

$$1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$$

We prove that (3.1) is true for  $n = k + 1$ , that is

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}$$

Obviously, since  $1 + 2 + 3 + \cdots + k = \frac{k(k+1)}{2}$  we have

$$1 + 2 + 3 + \cdots + k + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$$

So, (3.1) is true for  $n = k + 1$ . By the principle of induction, it follows that, (3.1) is true for any integer  $n \geq 1$ .  $\square$ **Example 2.** Prove that

$$1 + 3 + 5 + \cdots + (2n-1) = n^2 \quad (3.2)$$

for any integer  $n \geq 1$ .*Solution.* For  $n = 1$ , (3.2) is true, since

$$1 = 1^2$$

Suppose that (3.2) is true for  $n = k \geq 1$ , that is

$$1 + 3 + 5 + \cdots + (2k-1) = k^2$$

We want to prove that (3.2) is also true for  $n = k + 1$ , that is

$$1 + 3 + 5 + \cdots + (2k-1) + (2(k+1)-1) = (k+1)^2$$

Since  $1 + 3 + 5 + \cdots + (2k-1) = k^2$ , we have

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k-1) + (2(k+1)-1) &= k^2 + (2(k+1)-1) \\ &= k^2 + 2k + 1 \\ &= (k+1)^2 \end{aligned}$$

Therefore, the (3.2) is true for  $n = k + 1$ .By the principle of induction, it follows that, (3.2) is true for any integer  $n \geq 1$ .  $\square$

**Example 3.** Prove for  $n \geq 1$

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n+1)! - 1 \quad (3.3)$$

*Proof.* For  $n = 1$ , the left hand side is  $1 \cdot 1!$ . The right hand side is  $2! - 1$ . They are equal.

Suppose this holds

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + k \cdot k! = (k+1)! - 1$$

We need to prove

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + k \cdot k! + (k+1) \cdot (k+1)! = (k+2)! - 1$$

Consider the left hand side

$$\begin{aligned} 1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + k \cdot k! + (k+1) \cdot (k+1)! &= (k+1)! - 1 + ((k+1) \cdot (k+1)!) \\ &= (k+1)!((k+1) + 1) - 1 \\ &= (k+2)! - 1 \end{aligned}$$

By mathematical induction, we have  $1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots + n \cdot n! = (n+1)! - 1$  for any integer  $n \geq 1$ .  $\square$

**Example 4.** Show that

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} \leq 2\sqrt{n} \quad (3.4)$$

for all positive integer.

*Proof.* For  $n = 1$ , the inequality (3.4) is true since

$$1 \leq 2\sqrt{1}$$

Suppose that (3.4) is true for some integers  $n = k \geq 1$ , that is

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} \leq 2\sqrt{k}$$

We must to prove that (3.4) is also true for  $n = k+1$ , that is

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} \leq 2\sqrt{k+1}$$

Consider the left hand side

$$\begin{aligned} 1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{k}} + \frac{1}{\sqrt{k+1}} &\leq 2\sqrt{k} + \frac{1}{\sqrt{k+1}} \\ &= \frac{2\sqrt{k}\sqrt{k+1} + 1}{\sqrt{k+1}} \\ &\leq \frac{(k+k+1) + 1}{\sqrt{k+1}} \quad (\text{by AM-GM inequality}) \\ &= 2\sqrt{k+1} \end{aligned}$$

Therefore, the inequality (3.4) is also true for  $n = k+1$ .

By principle of mathematical induction, (3.4) is true for any integer  $n \geq 1$ .  $\square$

**Example 5.** Show that  $n! > 3^n$  for any integer  $n \geq 7$ .

*Proof.* For any integer  $n \geq 7$ , let  $P(n)$  be the statement that  $n! > 3^n$ .

For  $n = 7$ , the statement  $P(7)$  says that  $7! = 5040 > 3^7 = 2187$ , which is true.

Suppose that  $P(k)$  holds for some integer  $k \geq 7$ , that is

$$k! > 3^k.$$

It remains to show that  $P(k+1)$  holds, that is,

$$(k+1)! > 3^{k+1}$$

Obviously,

$$\begin{aligned} (k+1)! &= (k+1)k! > (k+1)3^k \\ &\geq (7+1)3^k = 8 \cdot 3^k \\ &> 3 \cdot 3^k = 3^{k+1} \end{aligned}$$

Therefore, the statement  $P(k+1)$  holds. Thus, by the principle of induction, it follows that  $n! > 3^n$  for all  $n \geq 7$ .  $\square$

**Example 6.** Prove by induction that  $n^3 - n$  is divisible by 3 for all natural number  $n$ .

*Proof.* For  $n = 1$ ,

$$\begin{aligned} n^3 - n &= 1 - 1 \\ &= 0 \end{aligned}$$

which is divisible by 3.

Assume that the statement is true for  $n = k$ , that is  $k^3 - k$  is divisible by 3. Now, we must prove that the statement is true for  $n = k+1$ , that is  $(k+1)^3 - (k+1)$  is divisible by 3.

Obviously, we have

$$\begin{aligned} (k+1)^3 - (k+1) &= k^3 + 3k^2 + 3k + 1 - k - 1 \\ &= (k^3 - k) + 3(k^2 + k) \end{aligned}$$

Since  $k^2 + k$  is an integer and  $k^3 - k$  is divisible by 3 (hypothesis induction), we have  $n^3 - n$  is divisible by 3 for  $n = k+1$ .

By mathematical induction  $n^3 - n$  is divisible by 3 for all positive integers  $n$ .  $\square$

**Example 7.** Prove by induction that  $11^n - 6$  is divisible by 5 for every integer  $n > 0$ .

*Proof.* Let  $P(n)$  be the mathematical statement:  $11^n - 6$  is divisible by 5.

When  $n = 1$  we have  $11^1 - 6 = 5$  which is divisible by 5. So  $P(1)$  is correct.

Assume that  $P(k)$  is correct for some integer  $k \geq 1$ . That means  $11^k - 6$  is divisible by 5 and hence  $11^k - 6 = 5m$  for some integer  $m$ . So  $11^k = 5m + 6$ .

We will now show that  $P(k+1)$  is correct. Always keep in mind what we are aiming for and what we know to be true. In this case we want to show that  $11^{k+1} - 6$  can be expressed as a multiple of 5, so we will start with the formula  $11^{k+1} - 6$  and we will rearrange it into something involving multiples of 5. At some point we will also want to use the assumption that  $11^k = 5m + 6$ .

$$\begin{aligned} 11^{k+1} - 6 &= 11 \cdot 11^k - 6 \\ &= 11(5m + 6) - 6 \\ &= 5(11m + 12) \end{aligned}$$

As  $11m + 12$  is an integer we have that  $11^{k+1} - 6$  is divisible by 5, so  $P(k+1)$  is correct. By principle of mathematical induction  $P(n)$  is correct for all positive integers  $n$ .  $\square$

**Example 8.** Prove that a set of  $n$  elements has  $2^n$  subsets.

*Proof.* Denote the number of subsets of a set of  $n$  elements by  $S(n)$ . Take a set of  $n$  elements. Take one of the elements, say  $\varepsilon$ , and consider separately the subsets which contain  $\varepsilon$  and those which do not.

There are  $S(n-1)$  subsets not containing  $\varepsilon$  (because they are the subsets of the  $(n-1)$  element set consisting of the original set without  $\varepsilon$ ).

The number of subsets which do contain  $\varepsilon$  is also  $S(n-1)$ , since a subset with  $\varepsilon$  is formed by adding  $\varepsilon$  to one of the subsets without  $\varepsilon$ .

So,

$$S(n) = S(n-1) + S(n-1) = 2S(n-1)$$

Similarly,

$$S(n-1) = 2S(n-2)$$

and so on, so that,

$$S(n) = 2S(n-1) = 4S(n-2) = \dots = 2^{n-1}S(1)$$

and,  $S(1) = 2$  (since a one-element set has two subsets, namely the whole set and the empty set).

Thus, the number of subsets of a set of  $n$  elements is  $S(n) = 2^n$ .  $\square$

**Example 9.** Prove that  $(3 + \sqrt{5})^n + (3 - \sqrt{5})^n$  is an even integer for all positive natural numbers  $n$ .

*Proof.* We will prove this example by strong induction<sup>1</sup>. Write

$$f(n) = \alpha^n + \beta^n$$

<sup>1</sup> Another way is using Newton's Binomial theorem.

where  $\alpha = 3 + \sqrt{5}$  and  $\beta = 3 - \sqrt{5}$ . It is straightforward to check that  $f(1) = 6$  and  $f(2) = 28$  are even integers.

Suppose  $f(k)$  and  $f(k+1)$  are both even integers for some positive integer  $k$ . Consider the case  $n = k+2$ .

Notice that  $\alpha$  and  $\beta$  are the roots of the equation

$$x^2 - 6x + 4 = 0.$$

So  $\alpha^2 = 6\alpha - 4$  and  $\beta^2 = 6\beta - 4$ , and thus

$$\begin{aligned} f(k+2) &= \alpha^{k+2} + \beta^{k+2} \\ &= \alpha^k(6\alpha - 4) + \beta^k(6\beta - 4) \\ &= 6(\alpha^{k+1} + \beta^{k+1}) - 4(\alpha^k + \beta^k) \\ &= 6f(k+1) - 4f(k) \end{aligned}$$

It follows that  $f(k+2)$  must also be an even integer.

By principle of mathematical induction, we conclude that  $f(n)$  is an even integer for all natural numbers  $n$ .  $\square$

**Example 10.** Prove that every positive integer greater than 1 can be written as a product of primes.

*Proof.* If  $n$  is a prime, then it is already a product of primes. If  $n$  is not a prime, then  $n = n_1 n_2$  where neither  $n_1$  or  $n_2$  are 1. So,  $n_1$  and  $n_2$  are both less than  $n$ .

We take as our induction hypothesis the statement, "Any number less than  $n$ , but greater than 1, can be written as a product of primes". So,  $n_1$  and  $n_2$  are products of primes and therefore  $n = n_1 n_2$  is also a product of primes.

So the truth of the result for all numbers less than  $n$  implies the truth of the result for  $n$ . The result is true for  $n = 2$ , so, by induction, it is true for all  $n > 1$ .  $\square$

### EXERCISE

**Exercise 1.** Prove by mathematical induction the following:

1.  $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$
2.  $1^3 + 2^3 + 3^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$
3.  $2 + 4 + 6 + \dots + 2n = n(n+1)$
4. For all positive integers  $n$ ,  $n^2 - n$  is even.

**Exercise 2.** Prove by induction that  $2^n > 2n$  for every positive integer  $n > 2$ .

**Exercise 3.** Use mathematical induction to prove that

$$\frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{n^2} < 1 - \frac{1}{n}$$

where  $n \geq 2$ .

**Exercise 4.** Use mathematical induction to prove that

$$\sum_{k=1}^n \frac{1}{k(k+2)} = \frac{3}{4} - \frac{2n+3}{2(n+1)(n+2)}$$

**Exercise 5.** Prove that

$$\frac{1}{4 \cdot 1^2 - 1} + \frac{1}{4 \cdot 2^2 - 1} + \frac{1}{4 \cdot 3^2 - 1} + \cdots + \frac{1}{4n^2 - 1} = \frac{n}{2n+1}$$

for every natural number  $n$ .

**Exercise 6.** Show that, the value of expression

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + \cdots + (-1)^{n-1} \cdot \frac{1}{n}$$

is always positive.

**Exercise 7.** Show that:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}} = 2 \cos \frac{\pi}{2^{n+1}}$$

where there are  $n$  nested square roots in the expression on the left.

**Exercise 8.** Let  $(a_n)$  be defined by  $a_1 = \sqrt{2}$  and  $a_{n+1} = \sqrt{2 + a_n}$  for  $n \in \mathbb{N}^*$ . Show that  $a_n \leq 2$  for all positive natural number  $n$ .

**Exercise 9.** Prove that the expression

1.  $3^{2n} - 1$  is divisible by 8 for any integer  $n \geq 0$ .
2.  $n^3 - n$  is divisible by 6 for all positive integers.
3.  $3^{3n+3} - 26n - 27$  is a multiple of 169 for every natural numbers  $n$ .
4.  $3^{2n+3} + 40n - 27$  is divisible by 64 for all positive integers  $n$ .

**Exercise 10\*.** (Fermat's Little Theorem) Prove that  $n^p - n$  is divisible by  $p$  for any prime  $p$ .

**Exercise 11.** Show that a number which consists of  $3^n$  equal digits is divisible by  $3^n$ .

**Exercise 12.** Prove that the internal angles of an  $n$ -sided polygon total  $(n \cdot 180^\circ - 360^\circ)$ .

**Exercise 13.** (Bernoulli's inequality<sup>2</sup>) Prove that

$$(1 + a)^n \geq 1 + na$$

for any positive integer  $n$ .

**Exercise 14.** Prove that  $2015^{2018} + 2016^{2018} < 2017^{2018}$

**Exercise 15.** For any natural number  $n$ ,

1. Calculate,

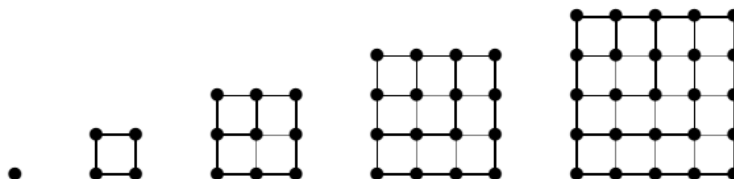
$$\frac{1}{2!} + \frac{1}{3!} + \frac{3}{4!} + \dots + \frac{n}{(n+1)!}$$

for a few small value of  $n$ .

2. Make a conjecture about the formula for this expression.

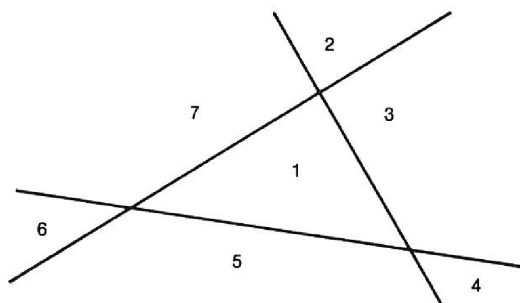
3. Prove your conjecture by mathematical induction.

**Exercise 16.** (Pentagonal numbers.) To define sequence of pentagonal numbers  $P_m(n)$  one needs to pack balls into regular  $m$ -gons with  $n$  balls on each side. For example,  $P_4(1) = 1, P_4(2) = 4, P_4(3) = 9, P_4(4) = 16, P_4(5) = 25$ . (Look at pictures below)



Establish the formula for  $P_m(n)$ , where  $m \geq 3$ .

**Exercise 17\*.** Suppose that we draw on a plane  $n$  line in "general position" (i.e. with no three concurrent, and no two parallel). Let  $s_n$  be the number of regions into which these lines divide the plane, for example,  $s_3 = 7$  in following figure.



1. By drawing the diagrams, find  $s_1, s_2, s_3, s_4$  and  $s_5$ .

<sup>2</sup>Jacob Bernoulli first published the inequality in his treatise "Positiones Arithmeticae de Seriebus Infinitis" (Basel, 1689), where he used the inequality often.

2. From these results, make a conjecture about formulas for  $s_n$ .

3. Prove this formula by mathematical induction.

**Exercise 18.** Prove that for all natural numbers  $n$ , there exist distinct integers  $x, y, z$  for which

$$x^2 + y^2 + z^2 = 14^n.$$

**Exercise 19\*.** Set  $S$  contains all positive integers from 1 to  $2n$ . Prove that among any  $n + 1$  numbers chosen from  $S$  there are two numbers such that one is a factor of the other.

**Exercise 20\*.** Prove that if  $x + \frac{1}{x}$  is integer then  $x^n + \frac{1}{x^n}$  is also integer for any positive integer  $n$ .

**Exercise 21\*.** (Chebyshev Polynomials) Define  $P_i(x)$  as follows:

$$P_0(x) = 1$$

$$P_1(x) = x$$

$$P_{n+1}(x) = xP_n(x) - P_{n-1}(x), \text{ for } n > 0.$$

Show that

$$P_n(2 \cos \theta) = \frac{\sin(n+1)\theta}{\sin \theta}$$

**Exercise 22\*.** (Quicksort) Prove the correctness of the following computer algorithm to sort a list of  $n$  numbers into ascending order. Assume that the original list is

$$\{x_0, x_1, \dots, x_{n-1}\}.$$

$\text{Sort}(j, k)$  where  $j \leq k$  sorts the elements from  $x_j$  to  $x_{k-1}$ . In other words, to sort the entire list of  $n$  elements, call  $\text{Sort}(0, n)$ . (Note that  $\text{Sort}(j, j)$  sorts an empty list.)

Here is the algorithm:

- **Case 1:** If  $k - j \leq 1$ , do nothing.
- **Case 2:** If  $k - j > 1$ , rearrange the elements from  $x_{j+1}$  through  $x_{k-1}$  so that the first slots in the list are filled with numbers smaller than  $x_j$ , then put in  $x_j$ , and then all the numbers larger than  $x_j$ . (This can be done by running a pointer from the  $(k-1)^{th}$  slot down and from the  $(j+1)^{th}$  slot up, swapping elements that are out of order. Then put  $x_j$  into the slot between the two lists.) After this rearrangement, suppose that  $x_j$  winds up in slot  $m$ , where  $j \leq m < k$ . Now apply  $\text{Sort}(j, m)$  and  $\text{Sort}(m+1, k)$ .

**Exercise 23\*.** (The ATM Machine) Suppose an ATM machine has only two dollar and five dollar bills. You can type in the amount you want, and it will figure out how to divide things up into the proper number of two's and five's. Prove that, The ATM machine can generate any output amount  $n \geq 4$ .

**Exercise 24\***. (Tower of Hanoi<sup>3</sup>) Suppose you have three posts and a stack of  $n$  disks, initially placed on one post with the largest disk on the bottom and each disk above it is smaller than the disk below. A legal move involves taking the top disk from one post and moving it so that it becomes the top disk on another post, but every move must place a disk either on an empty post, or on top of a disk larger than itself. Show that for every  $n$  there is a sequence of moves that will terminate with all the disks on a post different from the original one. How many moves are required for an initial stack of  $n$  disks?

**Exercise 25\***. (Newton's Binomial Theorem) Prove that

$$(a+b)^n = C_n^0 a^n + C_n^1 a^{n-1} b^1 + C_n^2 a^{n-2} b^2 + \dots + C_n^k a^{n-k} b^k + \dots + C_n^n b^n$$

for any integer  $n \geq 1$ .

**Exercise 26\***. (APMO 1999<sup>4</sup>) Let  $a_1, a_2, \dots, a_n$  be a sequence of real number satisfying  $a_{i+j} \leq a_i + a_j$  for all  $i, j = 1, 2, \dots$ . Prove that

$$a_1 + \frac{a_2}{2} + \frac{a_3}{3} + \dots + \frac{a_n}{n} \geq a_n$$

for each positive integer  $n$ .

**Exercise 27\***. (AM-GM inequality) Let  $a_1, a_2, \dots, a_n$  be positive numbers. Prove that

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}$$

for every integer number  $n \geq 1$ .

### 3.2.5 Sequences

A *sequence* can be regarded as a function whose domain is the set of natural numbers or some subset of it of the type  $\{1, 2, 3, \dots, n\}$ . Sometimes, we use the functional notation  $u(n)$  for  $u_n$ .

The *first term* of the sequence is  $u_1$ . The number of terms in the sequence is  $n$ . The *general term* of the sequence is  $u_n$ . This term is dependent on the value of  $n$ .

A *finite sequence* always has a finite number of terms. An *infinite sequence* has an infinite number of terms. Every term is followed by a new term.

Given any sequence  $(u_n)$ , we have the following

- We call the sequence increasing if  $u_n < u_{n+1}$  for every  $n$ .
- We call the sequence decreasing if  $u_n > u_{n+1}$  for every  $n$ .
- If  $(u_n)$  is an increasing sequence or  $(u_n)$  is a decreasing sequence we call it monotonic.

<sup>3</sup>This game was invented by the French mathematician Edouard Lucas (1842 – 1891) who did his most important work in the theory of prime numbers.

<sup>4</sup><http://cms.math.ca/Competitions/APMO/exam/apmo1999.html>

- If there exists a number  $m$  such that  $m \leq u_n$  for every  $n$  we say the sequence is bounded below. The number  $m$  is sometimes called a lower-bound for the sequence.
- If there exists a number  $M$  such that  $u_n \leq M$  for every  $n$  we say the sequence is bounded above. The number  $M$  is sometimes called an upper-bound for the sequence.
- If the sequence is both bounded below and bounded above we call the sequence bounded.

### EXAMPLE

**Example 11.** Write the first five terms of the sequence defined by the given explicit formula:

$$1. u_n = \frac{n}{2n-1}$$

$$3. u_n = (n+1) \cos n\pi$$

$$2. u_n = \frac{2n-1}{2^n}$$

$$4. u_n = \frac{\sin \frac{n\pi}{2}}{n+1}$$

**Example 12.** Write the first five terms of the sequence defined by the given recursive formula:

$$1. \begin{cases} u_1 = 2 \\ u_n = 2u_{n-1} - 1, \forall n \geq 2 \end{cases}$$

$$3. \begin{cases} u_1 = -2, u_2 = 2 \\ u_n = 2u_{n-1} + u_{n-2}, \forall n \geq 3 \end{cases}$$

$$2. \begin{cases} u_1 = -3 \\ u_{n+1} = 2u_n - 1, \forall n \geq 1 \end{cases}$$

$$4. \begin{cases} u_1 = 0, u_2 = 2, u_3 = 1 \\ u_n = u_{n-1} - u_{n-2} + u_{n-3}, \forall n \geq 4 \end{cases}$$

**Example 13.** Prove that the sequence  $(u_n)$  defined by

$$\begin{cases} u_1 = 1 \\ u_n = 2u_{n-1} + 3, \forall n \geq 2 \end{cases}$$

has the general term formula  $u_n = 2^{n+1} - 3$ .

*Solution.* We will prove the following formula

$$u_n = 2^{n+1} - 3, \forall n \geq 2 \quad (3.5)$$

by mathematical induction.

For  $n = 2$  we have  $u_2 = 2u_1 + 3 = 5 = 2^3 - 3$ . So the formula (3.5) is true.

Assume that the formula (3.5) is true for some integer  $n = k \geq 2$ , that is  $u_k = 2^{k+1} - 3$ .

We must prove the formula (3.5) is also true for  $n = k + 1$ , that is

$$u_{k+1} = 2^{k+2} - 3$$

By the recursive formula, we have

$$\begin{aligned} u_{k+1} &= 2u_k + 3 \\ &= 2(2^{k+1} - 3) + 3 \\ &= 2 \cdot 2^{k+1} - 6 + 3 \\ &= 2^{k+2} - 3 \end{aligned}$$

Hence, the formula (3.5) is true for  $n = k + 1$ .

By mathematical induction, the formula (3.5) is true for all positive integer  $n \geq 2$ .  $\square$

**Example 14.** Let  $(u_n)$  be a sequence of natural numbers such that  $u_1 = 5, u_2 = 13$  and  $u_{n+2} = 5u_{n+1} - 6u_n$  for all natural numbers  $n$ . Prove that  $u_n = 2^n + 3^n$  for all natural numbers  $n$ .

*Proof.* We first check that  $u_1 = 5 = 2^1 + 3^1$  and  $u_2 = 13 = 2^2 + 3^2$ .

Suppose that  $u_k = 2^k + 3^k$  and  $u_{k+1} = 2^{k+1} + 3^{k+1}$  for some natural number  $k$ .

Then

$$\begin{aligned} u_{k+1} &= 5u_{k+1} - 6u_k \\ &= 5(2^{k+1} + 3^{k+1}) - 6(2^k + 3^k) \\ &= 4 \cdot 2^k + 9 \cdot 3^k \\ &= 2^{k+2} + 3^{k+2} \end{aligned}$$

Hence, if the formula holds for  $n = k$  and  $n = k + 1$ , it also holds for  $n = k + 2$ .

By the principle of mathematical induction, we have  $u_n = 2^n + 3^n$  for all natural numbers  $n$ .  $\square$

**Example 15.** Determine whether the sequence  $u_n = \frac{2n-3}{3n+4}$  is increasing, decreasing, or not monotonic.

*Solution.* First we observe that  $(u_n)$  is not decreasing since  $u_1 = -\frac{1}{7} < \frac{1}{10} = u_2$ . So it is either increasing or not monotonic.

Now consider the definition. We know that  $(u_n)$  is increasing if  $u_n < u_{n+1}$  for all  $n \geq 1$ . In this instance,

$$\begin{aligned} u_n &= \frac{2n-3}{3n+4} \\ u_{n+1} &= \frac{2(n+1)-3}{3(n+1)+4} = \frac{2n-1}{3n+7} \end{aligned}$$

So we need to show that

$$u_n = \frac{2n-3}{3n+4} < \frac{2n-1}{3n+7} = u_{n+1}$$

for every  $n \in \mathbb{N}$ . That is, we need to show that

$$(2n-3)(3n+7) < (2n-1)(3n+4)$$

This last inequality follows since  $-21 < -4$ .

We conclude that  $u_n < u_{n+1}$  for every  $n \in \mathbb{N}$ .  $\square$

**Example 16.** Determine whether the sequence  $(u_n)$  is increasing, decreasing, or not monotonic:

$$u_n = n + \left(\frac{1}{2}\right)^n, n \in \mathbf{N}^*$$

*Solution.* For every  $n \in \mathbf{N}^*$  we have

$$u_{n+1} - u_n = \left(1 - \frac{1}{2^n}\right) + \frac{1}{2^{n+1}} > 0$$

That means  $u_{n+1} > u_n$  for every  $n \in \mathbf{N}^*$ . Therefore  $(u_n)$  is increasing.  $\square$

**Example 17.** Determine if the sequence  $(u_n)$  defined by  $u_n = -n^2$  is monotonic and/or bounded.

*Solution.* This sequence is a decreasing sequence (and hence monotonic) because,

$$-n^2 > -(n+1)^2$$

Also, since the sequence terms will be either zero or negative this sequence is bounded above. We can use any positive number or zero as the bound,  $M$ , however, it's standard to choose the smallest possible bound if we can and it's a nice number. So, we'll choose  $M = 0$  since,

$$-n^2 \leq 0 \forall n.$$

This sequence is not bounded below however since we can always get below any potential bound by taking  $n$  large enough. Therefore, while the sequence is bounded above it is not bounded.  $\square$

**Example 18.** Prove that the sequence  $(u_n)$  defined by the general term formula

$$u_n = \frac{2n-1}{n+1}$$

is bounded above by 2.

*Proof.* Consider the explicit formula of  $(u_n)$ :

$$\begin{aligned} u_n &= \frac{2n-1}{n+1} \\ &= 2 - \frac{3}{n+1} \end{aligned}$$

Since  $n \geq 1$  we have  $\frac{3}{n+1} > 0$ , and thus  $2 - \frac{3}{n+1} < 2$  for all natural numbers  $n$ . Therefore  $u_n < 2$  for all natural numbers  $n$ . Hence,  $(u_n)$  is bounded above by 2.  $\square$

**Example 19.** Prove that the sequence  $(u_n)$  defined by the general term formula

$$u_n = \frac{n+2}{2n-1}$$

is bounded below  $\frac{1}{2}$ .

*Proof.* Consider the explicit formula of  $(u_n)$ :

$$\begin{aligned} u_n &= \frac{n+2}{2n-1} \\ &= \frac{1}{2} + \frac{5}{4n+2} \end{aligned}$$

Since  $n \geq 1$  we have  $\frac{5}{4n+2} > 0$ , and thus  $\frac{1}{2} + \frac{5}{4n+2} > \frac{1}{2}$  for all natural numbers  $n$ . Therefore  $u_n > \frac{1}{2}$  for all natural numbers  $n$ . So,  $(u_n)$  is bounded below by  $\frac{1}{2}$ .  $\square$

**Example 20.** Determine whether the sequence  $(u_n)$  is bounded:

$$u_n = (-1)^n + \cos n, n \in \mathbf{N}^*$$

*Solution.* For every  $n \in \mathbf{N}^*$  we have

$$\begin{aligned} -1 &\leq \cos n \leq 1 \\ \Leftrightarrow -2 &\leq (-1)^n + \cos n \leq 2 \end{aligned}$$

Therefore  $(u_n)$  is bounded above by 2, and bounded below by  $-2$ . So, it is bounded.  $\square$

### EXERCISE

**Exercise 28.** Write the first five terms of the sequence defined by the given recursive or explicit formula:

1.  $u_n = \frac{2n^2-3}{n}$

3.  $u_n = (-1)^n \cdot \sqrt{4^n}$

2.  $u_n = \sin^2 \frac{n\pi}{4} + \cos \frac{2n\pi}{3}$

4.  $u_1 = 1, u_2 = 2, u_n = u_{n-1} + 2u_{n-2}$

**Exercise 29.** A Hailstone Sequence is a recursive sequence of positive integers in which  $a_n = \frac{1}{2}a_{n-1}$  if  $a_{n-1}$  is even, but  $a_n = 3a_{n-1} + 1$  if  $a_{n-1}$  is odd. Choosing different values of  $a_1$  results in different Hailstone Sequences. What is the smallest value of  $a_1$  that will generate 11 as part of its Hailstone Sequence?

**Exercise 30.** Determine the general term's formula of a sequence  $(u_n)$ , defined by the relations

$$u_1 = 3, u_{n+1} = 2u_n, \forall n \geq 1.$$

**Exercise 31.** Define a sequence of integers such that the first term of the sequence is 2, and the sequence is then continued by alternately multiplying by 2 and subtracting 1. The first several terms of the sequence, found in this way, are

$$2, 4, 3, 6, 5, 10, 9, 18, 17, 34, 33, 66, \dots$$

What is the value of the first term in the sequence which is greater than 2012?

**Exercise 32.** Determine whether the sequence  $(u_n)$  is increasing, decreasing, or not monotonic:

1.  $u_n = n^2 + n$

3.  $u_n = \sin \frac{n\pi}{3}$

2.  $u_n = \frac{1}{2n-5}$

4.  $u_n = 1 - \sqrt{n+1}$

**Exercise 33.** Determine whether the sequence  $(u_n)$  is increasing, decreasing, or not monotonic.

1.  $u_n = \frac{1}{2^n}$

3.  $u_n = \sqrt{n+1} - \sqrt{n}$

2.  $u_n = \frac{n}{2^n}$

4.  $u_n = \frac{2^n+4}{4^n}$

**Exercise 34.** Determine whether the sequence  $(u_n)$  is increasing, decreasing, or not monotonic.

1.  $u_1 = 3, u_{n+1} = u_n + 2$

2.  $u_1 = 1, u_{n+1} = 2u_n + 1$

**Exercise 35.** Show that the sequence  $(u_n)$  defined by

$$u_n = \frac{3n-2}{2n+3}$$

is bounded above by  $\frac{3}{2}$ .

**Exercise 36.** Prove that the sequence  $(u_n)$  defined by

$$u_n = \frac{3n+7}{n+1}$$

is bounded.

**Exercise 37\*.** Find the natural number  $n$  such that

$$\frac{1}{1.3} + \frac{1}{2.4} + \frac{1}{3.5} + \dots + \frac{1}{(2n-1)(2n+1)} + \frac{1}{2n(2n+2)} = \frac{14651}{19800}$$

**Exercise 38\*.** If a sequence  $(a_n)$  satisfies,

$$a_{n+1} = \frac{a_n}{a_n + 1}$$

Show that,

$$a_n = \frac{a_0}{na_0 + 1}.$$

**Exercise 39\*.** (ARML<sup>5</sup> 2011) Define the sequence of positive integers  $(a_n)$  as follows:

$$\begin{cases} a_1 = 1 \\ \text{for } n \geq 2, a_n \text{ is the smallest possible positive value of } n - a_k^2, \text{ for } 1 \leq k < n \end{cases}$$

For example,  $a_2 = 2 - 1^2 = 1$ , and  $a_3 = 3 - 1^2 = 2$ . Compute  $a_1 + a_2 + \dots + a_{50}$ .

<sup>5</sup>American Regions Mathematics League: <http://www.arml.com>

**Exercise 40\***. Consider the Fibonacci<sup>6</sup> sequence  $(u_n)$ , defined by the relation  $u_1 = u_2 = 1$ , and  $u_n = u_{n-1} + u_{n-2}$  for  $n \geq 3$ .

1. Compute  $u_{20}$ .

2. Prove that

$$u_n = \frac{1}{\sqrt{5}} \left( \left( \frac{1+\sqrt{5}}{2} \right)^n - \left( \frac{1-\sqrt{5}}{2} \right)^n \right)$$

3. Prove the formula  $u_{k+m} = u_{k-1}u_m + u_k u_{m+1}$

**Exercise 41\***. Let a sequence  $(u_n)$  be defined by  $u_1 = 0$  and  $u_{n+1} = 5u_n + \sqrt{24u_n^2 + 1}$  for  $n \geq 1$ . Prove that every term of this sequence is integer number.

**Exercise 42\***. (MathPrize 2013<sup>7</sup>) Let  $a_0, a_1, a_2, \dots$  be an infinite sequence of real numbers such that  $a_0 = \frac{4}{5}$  and

$$a_n = 2a_{n-1}^2 - 1$$

for every positive integer  $n$ . Let  $c$  be the smallest number such that for every positive integer  $n$ , the product of the first  $n$  terms satisfies the inequality

$$a_0 a_1 \dots a_{n-1} \leq \frac{c}{2^n}.$$

What is the value of  $100c$ , rounded to the nearest integer?

### 3.2.6 Arithmetic Sequence

A sequence  $(u_n)$  is called *arithmetic sequence* or *arithmetic progression* if

$$u_n = u_{n-1} + d, \forall n \geq 2$$

where  $u_1$  is called *the first term* and the constant  $d$  is called *the common difference* of the arithmetic sequence.

Here, we shall use the following notations for an arithmetic progression:

- The *general term* of an arithmetic sequence is  $u_n = u_1 + (n-1)d$
- The sum to  $n$  terms of arithmetic sequence is

$$S_n = u_1 + u_2 + \dots + u_n = \frac{n(u_1 + u_n)}{2} = \frac{1}{2}n(2u_1 + (n-1)d)$$

Simple property  $u_k = \frac{u_{k-1} + u_{k+1}}{2}$  for all  $k \geq 2$ .

<sup>6</sup>Leonardo Bonacci (c. 1170 – c. 1250) - known as Fibonacci - was an Italian mathematician, considered to be "the most talented Western mathematician of the Middle Ages".

<sup>7</sup><http://mathprize.atfoundation.org/archive/2013/index>

**EXAMPLE**

**Example 21.** Find the  $11^{th}$  term and the  $n^{th}$  term expression of the arithmetic sequence whose first term is 5 and whose common difference is 3.

*Solution.* We are given  $u_1 = 5$  and  $d = 3$ . Thus

$$u_n = u_1 + (n - 1)d = 5 + 3(n - 1) = 3n + 2$$

and the  $11^{th}$  term is  $u_{11} = 35$ . □

**Example 22.** An arithmetic sequence has a common difference equal to 10 and its  $6^{th}$  term is equal to 52. Find its  $15^{th}$  term?

*Solution.* We use the general term formula for the  $6^{th}$  term, which is known, to write

$$u_6 = 52 = u_1 + 10(6 - 1) \Leftrightarrow u_1 = 2$$

Now that we know the first term and the common difference, we use the general term formula to find the  $15^{th}$  term as follows:  $u_{15} = 2 + 10(15 - 1) = 142$  □

**Example 23.** A visual and performing arts group wants to hire a community events leader. The person will be paid \$12 for the first hour of work, \$19 for two hours of work, \$26 for three hours of work, and so on.

1. Write the general term that you could use to determine the pay for any number of hours worked.
2. What will the person get paid for 6h of work?

*Solution.* The common difference is  $d = 7$  and the general term is

$$u_n = 19 + 7(n - 1) = 12 + 7n.$$

He will get paid  $u_6 = \$54$  for 6h of work. □

**Example 24.** Many factors affect the growth of a child. Medical and health officials encourage parents to keep track of their child's growth. The general guideline for the growth in height of a child between the ages of 3 years and 10 years is an average increase of 5 cm per year. Suppose a child was 70 cm tall at age 3.

1. Write the general term that you could use to estimate what the child's height will be at any age between 3 and 10.
2. How tall is the child expected to be at age 10?

*Solution.* Setting  $h_n$  is the child's height will be at any age  $n$  between 3 and 10. Then, the general term is

$$h_n = 70 + 5(n - 3) = 5n + 55$$

The child will be  $5 \cdot 10 + 55 = 105$  cm tall at age 10.  $\square$

**Example 25.** Find the sum of all positive integers, from 5 to 1555 inclusive, that are divisible by 5?

*Solution.* The first few terms of a sequence of positive integers divisible by 5 is given by

$$5, 10, 15, 20, \dots$$

The above sequence has a first term equal to 5 and a common difference  $d = 5$ . We need to know the rank of the term 1555. We use the formula for the  $n$ th term as follows

$$1555 = u_1 + (n - 1)d \Leftrightarrow 1555 = 5 + 5(n - 1) \Leftrightarrow n = 311$$

Hence, we have  $S_{311} = \frac{311 \cdot (5 + 1555)}{2} = 242580$ .  $\square$

**Example 26.** Let the positive numbers  $a, b, c$  be an arithmetic sequences in that order. Prove that:

$$a^2 + 2bc = c^2 + 2ab$$

*Proof.* Because  $a, b, c$  form an arithmetic sequence, we have

$$\begin{aligned} b - a &= c - b \\ \Rightarrow (b - a)^2 &= (c - b)^2 \\ \Leftrightarrow a^2 - 2ab + b^2 &= c^2 - 2bc + b^2 \\ \Leftrightarrow a^2 + 2bc &= c^2 + 2ab \end{aligned}$$

That is the required claim.  $\square$

**Example 27.** Determine the first term of the arithmetic sequence  $(u_n)$  with common difference  $d$  in which:

$$\begin{cases} u_1 - u_3 + u_5 = 10 \\ u_1 + u_6 = 17 \end{cases}$$

*Solution.* Let  $d$  is common difference of  $(u_n)$  then  $u_3 = u_1 + 2d, u_5 = u_1 + 4d$  and  $u_6 = u_1 + 5d$ . Then, the given system of equations is equivalent to

$$\begin{cases} u_1 - (u_1 + 2d) + u_1 + 4d = 10 \\ u_1 + (u_1 + 5d) = 17 \end{cases} \Leftrightarrow \begin{cases} u_1 + 2d = 10 \\ 2u_1 + 5d = 17 \end{cases}$$

Solving this system of equations we obtain  $u_1 = -4$  and  $d = 7$ .  $\square$

**Example 28.** Let  $10 - 3x, 2x^2 + 3, 7 - 4x$  be an arithmetic sequences. Find  $x$ ?

*Solution.* Having  $10 - 3x, 2x^2 + 3, 7 - 4x$  form an arithmetic sequence iff

$$(10 - 3x) + (7 - 4x) = 2(2x^2 + 3)$$

Solving this equation we obtain  $x = \frac{15}{7}$ . □

**Example 29.** Find the sum of all terms from  $6^{th}$  term to the  $14^{th}$  term of an arithmetic sequence having the  $3^{rd}$  term is 16 and common difference is 4.

*Solution.* Since the common difference is 4 and the  $3^{rd}$  term is 16, we have the following equation:

$$16 = u_1 + 2 \cdot 4$$

Solving this equation we obtain  $u_1 = 8$ . Thus, the sum to  $6^{th}$  term of the given arithmetic sequence is

$$S_6 = \frac{6(8 \cdot 2 + 5 \cdot 5)}{2} = 123,$$

and the sum to  $14^{th}$  term of it is

$$S_{14} = \frac{14(8 \cdot 2 + 13 \cdot 5)}{2} = 567.$$

Therefore the sum of all terms from  $6^{th}$  term to the  $14^{th}$  term of the given arithmetic sequence is  $S = S_{14} - S_6 = 444$ . □

**Example 30.** The interior angles of a convex polygon form an arithmetic progression with a common difference of  $4^\circ$ . Determine the number of sides of the polygon if its largest interior angle is  $172^\circ$ .

*Solution.* Let  $n$  be the number of sides of the polygon. The first step to solving this problem is to determine the sum of the interior angles determined by the arithmetic progression in terms of  $n$ . It is fairly clear that the progression is  $172, 168, 164, \dots, 172 - 4(n - 1)$ . Thus we can find its sum to be:

$$\begin{aligned} & n \cdot \frac{172 + (172 - 4(n - 1))}{2} \\ &= 172n - 2n(n - 1) \\ &= 174n - 2n^2 \end{aligned}$$

We also know the formula for the sum of the interior angles of an  $n$ -sided polygon<sup>8</sup>:  $180(n - 2) = 180n - 360$ . Thus we can equate these two and solve for  $n$ :

$$\begin{aligned} 180n - 360 &= 174n - 2n^2 \\ \Leftrightarrow n^2 + 3n - 180 &= 0 \\ \Leftrightarrow n &= 12 \vee n = -15. \end{aligned}$$

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<sup>8</sup>Refer to exercise 12 at page 18

Obviously  $n = -15$  is an extraneous solution, and so we know that  $n = 12$ , and thus the polygon is a dodecagon.  $\square$

**Example 31.** Assume that  $x_1, x_2, x_3$  are three distinct roots of the equation

$$x^3 - 3x^2 - 9x + m = 0.$$

Determine  $m$  if  $x_1, x_2, x_3$  form an arithmetic sequence in that order?

*Solution.* Since  $x_1, x_2, x_3$  form an arithmetic sequence in that order, we have

$$x_1 + x_3 = 2x_2$$

Using the Viét's theorem, we have

$$x_1 + x_2 + x_3 = -\frac{b}{a} = 1$$

Therefore we have  $x_2 + 2x_2 = 3$ , giving  $x_2 = 1$ . Now, 1 is a root of the given equation, that is

$$1^3 - 3 \cdot 1^2 - 9 \cdot 1 + m = 0$$

Solving this equation we obtain  $m = 11$ . For  $m = 11$  the given equation is

$$x^3 - 3x^2 - 9x + 11 = 0$$

Solving this equation we obtain three roots  $1 - 2\sqrt{3}$ , 1 and  $1 + 2\sqrt{3}$ . It is easy to verify that these roots form an arithmetic sequence. Hence  $m = 11$  is the required value.  $\square$

**Exercise 43.** Let  $a_1, a_2, \dots, a_n$  be an arithmetic progression with common difference  $r$ . Find in terms of  $r$ , and  $a_1$  and  $a_n$  the explicit value of sum

$$S = \frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{n-1} a_n}$$

*Solution.* Note that

$$\frac{1}{a(a+r)} = \frac{1}{r} \left( \frac{1}{a} - \frac{1}{a+r} \right)$$

So  $S$  can be written as a "telescoping sum"<sup>9</sup> in which everything cancels except the first and the last term:

$$S = \frac{1}{r} \left( \left( \frac{1}{a_1} - \frac{1}{a_2} \right) + \left( \frac{1}{a_2} - \frac{1}{a_3} \right) + \dots + \left( \frac{1}{a_{n-1}} - \frac{1}{a_n} \right) \right) = \frac{1}{r} \left( \frac{1}{a_1} - \frac{1}{a_n} \right)$$

$\square$

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<sup>9</sup>Refer to <http://math.stackexchange.com/questions/13701/mathematical-telescoping>

**Example 32.** (Putnam competition<sup>10</sup> 1978) Let  $A$  be any set of 19 distinct integers chosen from the arithmetic progression  $1, 4, 7, \dots, 100$ . Prove that there must be two distinct integers in  $A$  whose sum is 104.

*Solution.* The pairs of distinct numbers in the arithmetic progression which add to 104 are:

$$(4, 100), (7, 97), \dots, (49, 55)$$

There are 16 of these pairs. The numbers 1 and 52 could be in  $A$ , but don't appear in any pairs. Apart from these, there are at least  $19 - 2 = 17$  numbers in  $A$  which all must come from one of these pairs. Since  $17 > 16$  and there are 16 pairs, by the *Pigeonhole Principle*<sup>11</sup>, two of these numbers in  $A$  must come from the same pair; so these numbers in  $A$  are distinct and add to 104.  $\square$

### EXERCISE

**Exercise 44.** Given the arithmetic sequence:  $u_1 = 124, u_2 = 117, u_3 = 110, u_4 = 103, \dots$

1. Write down the common difference of the sequence.
2. Calculate the sum of the first 50 terms of the sequence.
3. If  $u_k$  is the first term in the sequence that is negative. Find the value of  $k$ .

**Exercise 45.** Determine the first term of the arithmetic sequence in which the  $16^{th}$  term is 110 and the common difference is 7.

**Exercise 46.** Let  $(u_n)$  be an arithmetic sequence in which  $u_1 = 1$  and  $u_2 = 7$ .

1. Determine the common difference  $d$  of this arithmetic sequence
2. Determine  $u_2, u_3, u_{2015}$  and the general term.

**Exercise 47.** Find the arithmetic sequence  $(u_n)$  having the given values of

$$1. \begin{cases} u_3 = -15 \\ u_{14} = 18 \end{cases}$$

$$4. \begin{cases} u_{17} - u_{20} = 9 \\ u_{17}^2 + u_{20}^2 = 153 \end{cases}$$

$$2. \begin{cases} u_2 - u_3 + u_5 = 10 \\ u_4 + u_6 = 26 \end{cases}$$

$$5. \begin{cases} u_1 + 2u_5 = 0 \\ S_4 = 14 \end{cases}$$

$$3. \begin{cases} u_7 - u_3 = 8 \\ u_2 u_7 = 75 \end{cases}$$

$$6. \begin{cases} u_1 + u_3 + u_5 = -12 \\ u_1 u_2 u_3 = 8 \end{cases}$$

<sup>10</sup>[https://en.wikipedia.org/wiki/William\\_Lowell\\_Putnam\\_Mathematical\\_Competition](https://en.wikipedia.org/wiki/William_Lowell_Putnam_Mathematical_Competition)

<sup>11</sup>Refer to [https://en.wikipedia.org/wiki/Pigeonhole\\_principle](https://en.wikipedia.org/wiki/Pigeonhole_principle)

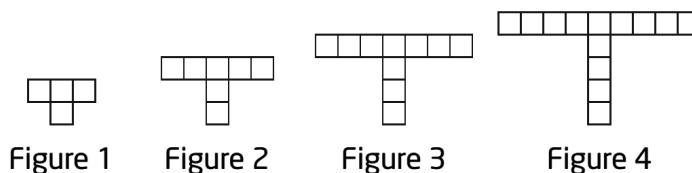
**Exercise 48.** The first term of an arithmetic sequence is 5, the last term is 45 and the sum of all terms is 400. Determine the common difference and the other terms?

**Exercise 49.** The terms  $5x + 2$ ,  $7x - 4$ , and  $10x + 6$  are consecutive terms of an arithmetic sequence. Determine the value of  $x$  and state the three terms.

**Exercise 50.** A single square is made from 4 matchsticks. Two squares in a row needs 7 matchsticks and 3 squares in a row needs 10 matchsticks. Determine: The first term, the common difference, the formula for the general term, how many matchsticks are in a row of 25 squares?



**Exercise 51.** Each square in this pattern has a side length of 1 unit. Assume the pattern continues.



1. Write an equation in which the perimeter is a function of the figure number.
2. Determine the perimeter of Figure 9.
3. Which figure has a perimeter of 76 units?

**Exercise 52.** Susan joined a fitness class at her local gym. Into her workout, she incorporated a sit-up routine that followed an arithmetic sequence. On the 6th day of the program, Susan performed 11 sit-ups. On the 15th day she did 29 sit-ups.

1. Write the general term that relates the number of sit-ups to the number of days.
2. If Susan's goal is to be able to do 100 sit-ups, on which day of her program will she accomplish this?

**Exercise 53.** Find the natural number  $n$  such that  $C_n^1, C_n^2, C_n^3$  form an arithmetic sequence in that order?

**Exercise 54.** Determine an arithmetic sequence having three terms, the sum of their terms is 9 and the sum of their squared term is 125.

**Exercise 55.** The side lengths of a quadrilateral produce an arithmetic sequence. If the longest side has a length of 24 cm and the perimeter is 60 cm, what are the other side lengths? Explain your reasoning.

**Exercise 56.** Earth has a daily rotation of  $360^\circ$ . One degree of rotation requires 4 min.

1. Write the sequence of the first five terms relating the number of minutes to the number of degrees of rotation.
2. Write an equation that describes this sequence.
3. Determine the time taken for a rotation of  $80^\circ$ .

**Exercise 57.** Determine the relationship between numbers  $m$  and  $n$  such that the equation

$$x^3 + 3x^2 - (24 + m)x - 26 - n = 0$$

has three roots, which form an arithmetic sequence.

**Exercise 58.** Find  $m$  such that the equation

$$x^4 - (3m + 5)x^2 + (m + 1)^2 = 0$$

has 4 roots, which form an arithmetic sequence.

**Exercise 59.** Given that  $a, b$  and  $c$  are successive terms in an arithmetic sequence. Calculate  $x$  if

$$(b - c)x^2 + (c - a)x + (a - b) = 0$$

**Exercise 60.** Let  $a, b, c$  be positive numbers and assume that  $a^2, b^2, c^2$  form an arithmetic sequence, prove that

$$\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$$

also form an arithmetic sequence.

**Exercise 61.** Let  $a, b, c$  be positive numbers. Prove that,  $a, b, c$  form an arithmetic sequence if and only if

$$\frac{1}{\sqrt{b} + \sqrt{c}}, \frac{1}{\sqrt{c} + \sqrt{a}}, \frac{1}{\sqrt{a} + \sqrt{b}}$$

also form an arithmetic sequence.

**Exercise 62\*.** Suppose that  $a_1, a_2, \dots, a_n$  form an arithmetic sequence, where  $a_i > 0$  for all  $i$ . Prove that

$$\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \dots + \frac{1}{a_{n-1} a_n} = \frac{n-1}{a_1 a_n}$$

**Exercise 63\*.** Prove that  $\sqrt{2}, \sqrt{3}, \sqrt{5}$  can't be the terms in any arithmetic sequence.

**Exercise 64\*.** Prove that if an infinite arithmetic progression of positive integers contains a perfect square, then it contains an infinite number of perfect squares.

**Exercise 65\*.** Prove that there are no arithmetic progressions of positive integers whose terms are all perfect squares.

**Exercise 66\*.** Compute

$$S = 1^2 - 2^2 + 3^2 - 4^2 + \cdots + (2n-1)^2 - (2n)^2$$

where  $n$  is a natural number.

### 3.2.7 Geometric Sequence

A sequence  $(u_n)$  is called *geometric sequence* or *geometric progression* if

$$u_{n+1} = u_n q, \forall n \geq 2$$

where  $u_1$  is called *the first term* and the constant  $q$  is called *the common ratio* of the geometric sequence.

Here, we shall use the following notations for an arithmetic progression:

- The *general term* of an geometric sequence is  $u_n = u_1 \cdot q^{n-1}$
- The sum to  $n$  terms of geometric sequence is

$$S_n = u_1 + u_2 + \cdots + u_n = u_1 \frac{1 - q^n}{1 - q}$$

Simple property  $u_k^2 = u_{k-1} \cdot u_{k+1}$  for all  $k \geq 2$ .

#### EXAMPLE

**Example 33.** In a geometric sequence, the second term is 28 and the fifth term is 1792. Determine the values of  $u_1$  and  $q$ , and list the first three terms of the sequence?

*Solution.* Since the second term is 28 and the fifth term is 1792 we have the following system of equations:

$$\begin{cases} u_1 \cdot q &= 28 \\ u_1 \cdot q^4 &= 1792 \end{cases}$$

Now divide the second equation by the first to obtain  $q^3 = 64 \Leftrightarrow q = 4$ . Substituting back into the first equation yields  $u_1 = 7$ . Hence the first three terms of the sequence is 7, 28, 112.  $\square$

**Example 34.** How many terms of the geometric sequence  $3, \frac{3}{2}, \frac{3}{4}, \dots$  are needed to give the sum  $\frac{3069}{512}$ ?

*Solution.* The given sequence is a arithmetic sequence where the first term  $u_1 = 3$  and common ratio  $q = \frac{1}{2}$ . Let  $n$  be the number of terms are needed to give the sum  $\frac{3069}{512}$ , then

$$\begin{aligned} S_n &= u_1 \cdot \frac{1 - q^n}{1 - q} \\ \Leftrightarrow \frac{3069}{512} &= 3 \cdot \frac{1 - (\frac{1}{2})^n}{1 - \frac{1}{2}} \end{aligned}$$

Solving this equation we obtain  $n = 10$ .  $\square$

**Example 35.** The sum of first three terms of a geometric is  $\frac{13}{12}$  and their product is  $-1$ . Find the common ratio and the terms.

*Solution.* Let  $\frac{u_1}{q}, u_1, u_1q$  be the first three terms of a geometric sequence. Then

$$\begin{cases} \frac{u_1}{q} + u_1 + u_1q = \frac{13}{12} \\ \frac{u_1}{q} \cdot u_1 \cdot u_1q = -1 \end{cases}$$

From the second equation, we get  $u_1^3 = -1$ , which yields  $u_1 = -1$ . Substituting back  $u_1 = -1$  into the first equation we have  $-\frac{1}{q} - 1 - q = \frac{13}{12}$ . Solving this equation we obtain  $q = -\frac{3}{4}$  or  $\frac{4}{3}$ . Thus, the first three terms of geometric sequence are  $\frac{4}{3}, -1, \frac{3}{4}$  for  $q = -\frac{3}{4}$  and  $\frac{3}{4}, -1, \frac{4}{3}$  for  $q = \frac{4}{3}$ .  $\square$

**Example 36.** A person has 2 parents, 4 grandparents, 8 great grandparents, and so on. Find the number of his ancestors during the ten generations preceding his own.

*Solution.* Here  $u_1 = 2, q = 2$  and  $n = 10$ . Using the sum formula

$$S_n = u_1 \frac{q^n - 1}{q - 1}$$

We have  $S_{10} = 2(2^{10} - 1) = 2046$ .

Hence, the number of ancestors preceding the person is 2046.  $\square$

**Example 37.** Insert three numbers between 1 and 256 so that the resulting sequence is a geometric sequence.

*Solution.* Let  $u_2, u_3, u_4$  be three numbers between 1 and 256 such that  $1, u_2, u_3, u_4, 256$  is a geometric sequence where the first term  $u_1 = 1$  and the last term  $u_5 = 256$ . Therefore  $256 = q^4$  giving  $q = \pm 4$ .

- For  $q = 4$ , we have  $x = u_1q = 4, y = u_1q^2 = 16, z = u_1q^3 = 64$ .
- Similarly, for  $q = -4$ , numbers are  $-4, 16$  and  $-64$ .

Hence, we can insert 4, 16, 64 or  $-4, 16, -64$  between 1 and 256 so that the resulting sequences are in geometric sequence.  $\square$

**Example 38.** Let  $(u_n)$  be a sequence defined by  $u_n = \frac{5}{2}$  and  $u_{n+1} = 3u_n - 1$  for all  $n \geq 1$ . Prove that the sequence  $(v_n)$  defined by  $v_n = u_n - \frac{1}{2}$  for all  $n \geq 1$  is a geometric sequence. Determine the first term and common ratio of  $(v_n)$ .

*Solution.* Since the definition of  $(u_n)$  and  $(v_n)$ , we have

$$v_{n+1} = u_{n+1} - \frac{1}{2} = 3u_n - 1 - \frac{1}{2} = 3\left(u_n - \frac{1}{2}\right) = 3v_n \text{ for all } n \geq 1.$$

Obviously,  $(v_n)$  is an geometric sequence where the first term  $v_1 = 2$  and common ratio  $q = 3$ .  $\square$

**Example 39.** Find the first term and common ratio of a geometric sequence  $(u_n)$  if

$$\begin{cases} u_4 - u_2 = 72 \\ u_5 - u_3 = 144 \end{cases}$$

*Solution.* Let  $u_1$  be the first term and  $q$  be the common ratio of the given sequence. Then,  $u_2 = u_1q$ ,  $u_3 = u_1q^2$ ,  $u_4 = u_1q^3$  and  $u_5 = u_1q^4$ . So, we can write the given system of equations as:

$$\begin{cases} u_1q^3 - u_1q = 72 \\ u_1q^4 - u_1q^2 = 144 \end{cases} \Leftrightarrow \begin{cases} u_1q(q^2 - 1) = 72 \\ q \cdot (u_1q(q^2 - 1)) = 144 \end{cases} \quad (3.6)$$

Substituting  $u_1q(q^2 - 1) = 72$  into the second equation we obtain  $72q = 144 \Leftrightarrow q = 2$ . Substituting back into the first equation yields  $u_1 = 12$ .  $\square$

**Example 40.** Determine the positive numbers  $a, b$  if the terms  $a, a + 2b, 2a + b$  are consecutive terms of an arithmetic sequence, and the terms  $(b + 1)^2, ab + 5, (a + 1)^2$  are consecutive terms of a geometric sequence.

*Solution.* It follows from  $a, a + 2b, 2a + b$  are consecutive terms of an arithmetic sequence and  $(b + 1)^2, ab + 5, (a + 1)^2$  are consecutive terms of a geometric sequence that

$$\begin{cases} a + (2a + b) = 2(a + 2b) \\ (b + 1)^2 \cdot (a + 1)^2 = (ab + 5)^2 \end{cases}$$

From the first equation we have  $a = 3b$ . Substituting it into the second equation yields

$$(b + 1)^2 \cdot (3b + 1)^2 = (3b \cdot b + 5)^2$$

Solving this equation we obtain  $b = 1$ , and then  $a = 3$ .  $\square$

**Example 41.** Suppose that the  $x$ -intercepts of the curve  $y = x^3 - (3m + 1)x^2 + (5m - 4)x - 8$  are distinct and they form a geometric sequence. Find  $m$ ?

*Solution.* Assume that the  $x$ -intercepts of the given curve are  $x_1, x_2, x_3$ . Then they are the roots of the following equation

$$x^3 - (3m + 1)x^2 + (5m - 4)x - 8 = 0 \quad (3.7)$$

and using Viét's theorem, we have

$$x_1 \cdot x_2 \cdot x_3 = 8$$

Notice that  $x_1, x_2, x_3$  form a geometric sequence in that order iff

$$x_2^2 = x_1 \cdot x_3$$

Therefore we have  $x_2^3 = 8$ , giving  $x_2 = 2$ . Now, 2 is a root of the given equation, that is

$$2^3 - (3m + 1) \cdot 2^2 + (5m - 4) \cdot 2 - 8 = 0$$

Solving this equation we obtain  $m = -6$ . For  $m = -6$  the equation (3.7) becomes

$$x^3 + 17x^2 - 34x - 8 = 0$$

Solving this equation we obtain three roots  $\frac{-19-\sqrt{345}}{2}$ , 2 and  $\frac{-19+\sqrt{345}}{2}$ . It is easy to verify that these roots form a geometric sequence. Hence  $m = -6$  is the required value.  $\square$

**Example 42.** Compute

$$S = 1 + \frac{1}{3} + \frac{1}{3^2} + \cdots + \frac{1}{3^{2015}}$$

*Solution.* This is the sum to the  $2016^{th}$  terms of a geometric sequence where the first term  $u_1 = 1$  and common ratio  $q = \frac{1}{3}$ . Thus,

$$S = u_1 \cdot \frac{1 - q^n}{1 - q} = 1 \cdot \frac{1 - (\frac{1}{3})^{2016}}{1 - \frac{1}{3}}$$

Therefore, we obtain  $S = \frac{3^{2016} - 1}{2 \cdot 3^{2015}}$ .  $\square$

**Example 43.** Find the sum of the sequence 7, 77, 777, 7777, ... to  $n$  terms?

*Solution.* This is not a geometric sequence, however, we can relate it to a geometric sequence by writing the terms as

$$\begin{aligned} S_n &= 7 + 77 + 777 + 7777 + \cdots \\ &= \frac{7}{9} (9 + 99 + 999 + 9999 + \cdots) \\ &= \frac{7}{9} ((10 - 1) + (10^2 - 1) + (10^3 - 1) + (10^4 - 1) + \cdots + (10^n - 1)) \\ &= \frac{7}{9} \left( \frac{10(10^n - 1)}{10 - 1} - n \right) \end{aligned}$$

Then the required sum is  $\frac{7}{9} \left( \frac{10(10^n - 1)}{9} - n \right)$ .  $\square$

### EXERCISE

**Exercise 67.** The first term of an arithmetic sequence is 0 and the common difference is 12.

1. Find the value of the 96th term of the sequence.
2. The first term of a geometric sequence is 6. The 6th term of the geometric sequence is equal to the 17th term of the arithmetic sequence given above.
  - (a) Write down an equation using this information.

(b) Calculate the common ratio of the geometric sequence.

**Exercise 68.** Prove that a sequence  $(a_n)$  defined by  $a_n = 2 \cdot 3^n$  forms a geometric sequence. Calculate the sum of its first eight terms?

**Exercise 69.** Suppose that  $a, b, c, d$  form a geometric sequence. Evaluate:

$$(a - c)^2 + (b - c)^2 + (b - d)^2 - (a - d)^2$$

**Exercise 70.** If  $a, b, a + b$ , and  $ab$  are positive numbers that form 4 consecutive terms in a geometric sequence, find  $a$ .

**Exercise 71.** A geometric sequence has 5 terms, common ratio  $q$  is equal to a quarter of first term, sum of their two first terms 24. Tìm cấp số nhân đó?

**Exercise 72.** Find the first term and common ratio of a geometric sequence  $(u_n)$  if:

1.  $u_5 = 96$  and  $u_9 = 192$
2.  $u_3 + u_5 = 90$  and  $u_2 - u_6 = 240$
3.  $6u_2 + u_5 = 1$  and  $3u_3 + 2u_4 = -1$
4.  $u_{20} = 8u_{17}$  and  $u_3 + u_5 = 272$

**Exercise 73.** The sum of first three terms of a geometric sequence is  $\frac{39}{10}$  and their product is 1. Find the common ratio and the terms.

**Exercise 74.** How many terms of geometric sequence  $3, 3^2, 3^3, \dots$  are needed to give the sum 120?

**Exercise 75.** The sum of first three terms of a geometric sequence is 16 and the sum of the next three terms is 128. Determine the first term, the common ratio and the sum to  $n^{\text{th}}$  terms of the geometric sequence.

**Exercise 76.** Given a geometric sequence where  $u_1 = 729$  and 7th term 64, determine  $S_7$ .

**Exercise 77.** Find a geometric sequence for which sum of the first two terms is -4 and the fifth term is 4 times the third term.

**Exercise 78.** If the 4th, 10th and 16th terms of a geometric sequence are  $x, y$  and  $z$ , respectively. Prove that  $x, y, z$  are in geometric sequence.

**Exercise 79.** A right triangle  $ABC$  has  $\hat{A} = 90^\circ$  and the length of three sides are  $a, b, c$  form a geometric sequence. Find the common ratio of it?

**Exercise 80.** Find the sum of the products of the corresponding terms of the sequences  $2, 4, 8, 16, 32$  and  $128, 32, 8, 2, \frac{1}{2}$ .

**Exercise 81.** Find four numbers forming a geometric progression in which the third term is greater than the first term by 9, and the second term is greater than the 4th by 18.

**Exercise 82.** If the  $p^{th}$ ,  $q^{th}$  and  $r^{th}$  terms of a gp are  $a, b$  and  $c$ , respectively. Prove that

$$a^{q-r} b^{r-p} c^{p-q} = 1$$

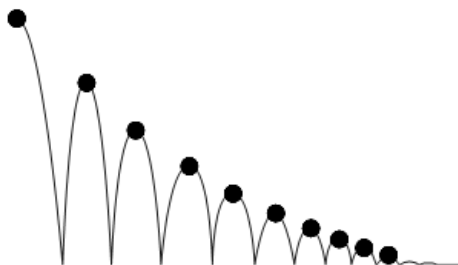
**Exercise 83.** If the first and the  $n^{th}$  term of a geometric sequence are  $a$  and  $b$ , respectively, and if  $P$  is the product of  $n$  terms, prove that  $P^2 = (ab)^n$ .

**Exercise 84.** Show that the ratio of the sum of first  $n$  terms of a geometric sequence to the sum of terms from  $(n+1)^{th}$  to  $(2n)^{th}$  term is  $\frac{1}{r^n}$ .

**Exercise 85.** The primary function for our kidneys is to filter our blood to remove any impurities. Doctors take this into account when prescribing the dosage and frequency of medicine. A person's kidneys filter out 18% of a particular medicine every two hours.

1. How much of the medicine remains after 12 h if the initial dosage was 250 mL? Express your answer to the nearest tenth of a millilitre.
2. When there is less than 20 mL left in the body, the medicine becomes ineffective and another dosage is needed. After how many hours would this happen?

**Exercise 86.** A ball is dropped from a height of 3.0 m. After each bounce it rises to 75% of its previous height.



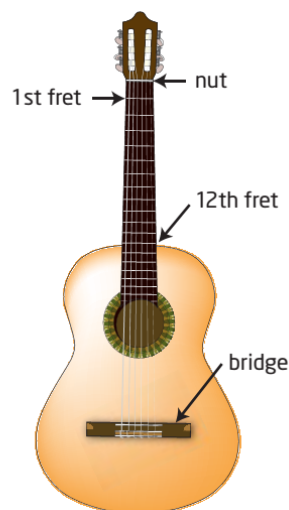
1. Write the first term and the common ratio of the geometric sequence. Write the general term for this sequence?
2. What height does the ball reach after the 6th bounce?
3. After how many bounces will the ball reach a height of approximately 40 cm?

**Exercise 87.** Bread and bread products have been part of our diet for centuries. To help bread rise, yeast is added to the dough. Yeast is a living unicellular micro-organism about one hundredth of a millimetre in size. Yeast multiplies by a biochemical process called budding. After mitosis and cell division, one cell results in two cells with exactly the same characteristics.



1. Write a sequence for the first six terms that describes the cell growth of yeast, beginning with a single cell.
2. Write the general term for the growth of yeast.
3. How many cells would there be after 25 doubling periods?
4. What assumptions would you make for the number of cells after 25 doubling periods?

**Exercise 88.** On a six-string guitar, the distance from the nut to the bridge is 38 cm. The distance from the first fret to the bridge is 35.87 cm, and the distance from the second fret to the bridge is 33.86 cm. This pattern approximates a geometric sequence.

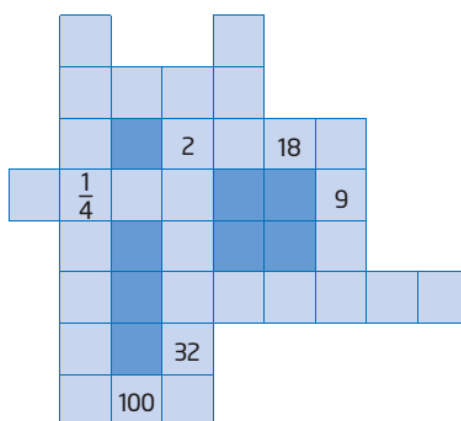


1. What is the distance from the 8th fret to the bridge?
2. What is the distance from the 12th fret to the bridge?
3. Determine the distance from the nut to the first fret.
4. Determine the distance from the first fret to the second fret.
5. Write the sequence for the first three terms of the distances between the frets. Is this sequence geometric or arithmetic? What is the common ratio or common difference?

**Exercise 89.** The Russian nesting doll or Matryoshka had its beginnings in 1890. The dolls are made so that the smallest doll fits inside a larger one, which fits inside a larger one, and so on, until all the dolls are hidden inside the largest doll. In a set of 50 dolls, the tallest doll is 60 cm and the smallest is 1 cm. If the decrease in doll size approximates a geometric sequence, determine the common ratio. Express your answer to three decimal places?



**Exercise 90.** Copy the puzzle. Fill in the empty boxes with positive numbers so that each row and column forms a geometric sequence.



**Exercise 91\*.** In a triangle the side lengths form a geometric sequence. Prove that this triangle has two angles measuring less than  $60^\circ$ .

**Exercise 92.** These numbers  $x, 3, y$  is a geometric sequence and  $x^4 = y\sqrt{3}$ . Find  $x, y$  and the common ratio  $q$  of that geometric sequence?

**Exercise 93.** Demonstrate that  $6^a, 6^b, 6^c$  forms a geometric sequence when  $a, b, c$  forms an arithmetic sequence.

**Exercise 94.** If  $x + 2, 2x + 1$ , and  $4x - 3$  are three consecutive terms of a geometric sequence, determine the value of the common ratio and the three given terms.

**Exercise 95\*.** Determine three terms of a geometric sequence if sum of them is 93 and, we can arrange them as (in that order) the first term, the second term and the seventh term of an arithmetic sequence.

**Exercise 96.** Calculate the value of these expressions

1.  $1 + 4 + 16 + \dots + 65536$
2.  $3 - 15 + 75 - \dots - 234375$
3.  $1 + 4 \cdot 2 + 7 \cdot 2^2 + 10 \cdot 2^3 + \dots + 288 \cdot 2^{99}$
4.  $1 + 2 \cdot 3 + 3 \cdot 3^2 + \dots + 100 \cdot 3^{99}$

**Exercise 97\*.** Compute

$$\left(2 + \frac{1}{2}\right)^2 + \left(4 + \frac{1}{4}\right)^2 + \dots + \left(2^n + \frac{1}{2^n}\right)^2$$

**Exercise 98\*.** Compute:  $3 + 33 + 333 + \dots + \underbrace{333\dots3}_{n \text{ digits } 3}$

**Exercise 99.** Find the sum to  $n$  terms of the sequence  $8, 88, 888, 8888, \dots$

### 3.2.8 Review

**Exercise 100.** Determine the sum of all the multiples of 4 between 1 and 999.

**Exercise 101.** It's About Time, in Langley, British Columbia, is Canada's largest custom clock manufacturer. They have a grandfather clock that, on the hours, chimes the number of times that corresponds to the time of day. For example, at 4:00 p.m., it chimes 4 times. How many times does the clock chime in a 24-h period?

**Exercise 102.** The second and fifth terms of an arithmetic sequence are 40 and 121, respectively. Determine the sum of the first 25 terms of the sequence.

**Exercise 103.** If  $x, 4, y$  are successive terms in an arithmetic sequence and  $x, 3, y$  are successive terms in a geometric sequence, calculate  $\frac{1}{x} + \frac{1}{y}$ .

**Exercise 104.** The sum of the first five terms of an arithmetic sequence is 85. The sum of the first six terms is 123. What are the first four terms of the sequence.

**Exercise 105.** Three different numbers, whose product is 125, are 3 consecutive terms in a geometric sequence. At the same time they are the first, third and sixth terms of an arithmetic sequence. Find the numbers.

**Exercise 106.** (ARML 2011) If  $1, x, y$  is a geometric sequence and  $x, y, 3$  is an arithmetic sequence, compute the maximum value of  $x + y$ .

**Exercise 107.** The sum of three consecutive terms in an arithmetic sequence is 2, and the sum of them squared is  $\frac{14}{9}$ . Determine these number?

**Exercise 108.** (Euclid Contest 2008) The numbers  $a, b, c$ , in that order, form a three term arithmetic sequence and  $a + b + c = 60$ . The numbers  $a - 2, b, c + 3$ , in that order, form a three term geometric sequence. Determine all possible values of  $a, b$  and  $c$ .

**Exercise 109.** These number  $x + 6y, 5x + 2y, 8x + y$  form an arithmetic sequence in that order, and  $x + \frac{5}{3}, y - 1, 2x - 3y$  form a geometric sequence in that order. Find  $x$  and  $y$ ?

**Exercise 110.** Suppose that  $5x - y, 2x + 3y, x + 2y$  form an arithmetic sequence, and  $(y + 1)^2, xy + 1, (x - 1)^2$  form an geometric sequence. Find the value of  $x, y$ ?

**Exercise 111.** If the interior angles of a pentagon form an arithmetic sequence and one interior angle is  $90^\circ$ , find all possible values of the largest angle in the pentagon.

**Exercise 112.** Find the 4 integers  $a, b, c$  and  $d$  that satisfy the following conditions:

- the sum of  $b$  and  $c$  is 30
- the sum of  $a$  and  $d$  is 35
- the numbers  $a < b < c < d$  are in geometric sequence
- the sum of the squares of the 4 numbers is 1261.

**Exercise 113.** How many terms in the arithmetic sequence  $7, 14, 21, \dots$  are between 40 and 28001?

**Exercise 114.** Suppose that three roots of equation  $x^3 - 3x^2 + mx + 3 - 2m = 0$  form an arithmetic sequence, find  $m$ ?

**Exercise 115.** Let  $ABC$  is right triangle which has  $\hat{A} = 90^\circ$ . Suppose that  $a, \frac{\sqrt{6}}{3} \times b, c$  form a geometric sequence in that order. Find the value of  $\hat{B}, \hat{C}$ ?

**Exercise 116.** A geometric sequence has 6 terms, the sum of first 5 terms is 31 and the sum of last 5 terms is 62. Determine this geometric sequence?

**Exercise 117\*.** Find the first 4 term of a geometric sequence if the sum of first 3 terms is  $16\frac{4}{9}$ , and in that order, they are first term, 4th term and 8th term of an arithmetic sequence.

**Exercise 118.** For the family of lines with equations of the form  $px + qy = r$ , and which all pass through the point  $(-1, 2)$ , prove that  $p, q$ , and  $r$  are consecutive terms of an arithmetic sequence.

**Exercise 119.** If  $f$  is a function such that  $f(1) = 2$  and  $f(n + 1) = \frac{3f(n) + 1}{3}$  for  $n = 1, 2, 3, \dots$ , what is the value of  $f(100)$ ?

**Exercise 120\*.** Prove that

$$\frac{1.3.5 \dots (2n-1)}{2.4.6.2n} < \frac{1}{\sqrt{2n+1}}$$

for every  $n \in \mathbb{N}$ .

**Exercise 121\*.** Demonstrate that

$$\frac{a^n + b^n}{2} \geq \left( \frac{a+b}{2} \right)^n$$

where  $a, b$  is positive numbers and  $n \in \mathbb{N}^*$ .

**Exercise 122.** Let  $f$  be a function which satisfies  $f(x+y) \geq f(x)f(y)$  for every pair number  $x, y$ . Prove that

$$f(x) \geq \left(f\left(\frac{x}{2^n}\right)\right)^{2^n}$$

for any real number  $x$  and any  $n \in \mathbf{N}^*$ .

**Exercise 123.** Prove that

$$\left(1 + \frac{1}{n}\right)^n < n$$

for every integer number  $n > 2$ .

**Exercise 124.** Prove that  $|\sin n\theta| \leq n|\sin \theta|$  for every natural number  $n$ .

**Exercise 125\*.** (ARML 2010) Let  $a_1, a_2, a_3, \dots$  be an arithmetic sequence, and let  $b_1, b_2, b_3, \dots$  be a geometric sequence. The sequence  $c_1, c_2, c_3, \dots$  has  $c_n = a_n + b_n$  for each positive integer  $n$ . If  $c_1 = 1, c_2 = 4, c_3 = 15$ , and  $c_4 = 2$ , compute  $c_5$ .

**Exercise 126\*.** (2011 Euclid Contest) Three different numbers are chosen at random from the set  $\{1, 2, 3, 4, 5\}$ . The numbers are arranged in increasing order. What is the probability that the resulting sequence is an arithmetic sequence?

### 3.3 Hint for some exercises

7. For  $n = 1$  we have:

$$\sqrt{2} = 2 \cos \frac{\pi}{2^2} = 2 \cos \frac{\pi}{4} = \sqrt{2}$$

Now assume it's true for  $k$  nested square roots:

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots + \sqrt{2}}}} = 2 \cos \frac{\pi}{2^{k+1}}$$

If we add 2 to both sides and take the square root, the left hand side will now have  $k + 1$  nested square roots, and the right hand side will be:

$$\sqrt{2 + 2 \cos \frac{\pi}{2^{k+1}}}$$

We just need to show that the value above is equal to

$$2 \cos \frac{\pi}{2^{k+2}}$$

14. By Bernoulli's inequality:

$$\left(\frac{2017}{2016}\right)^{2018} = \left(1 + \frac{1}{2016}\right)^{2018} \geq 1 + \frac{2018}{2016} > 2$$

Therefore

$$2017^{2018} > 2 \cdot 2016^{2018} > 2015^{2018} + 2016^{2018}$$

- 18.** For  $n = 1$  and  $n = 2$ , such integers exist as  $1^2 + 2^2 + 3^2 = 14$  and  $4^2 + 6^2 + 12^2 = 14^2$ .

Suppose for  $n = k$  (where  $k$  is some positive integer), such integers exist, i.e.  $x_0^2 + y_0^2 + z_0^2 = 14^k$  for some distinct integers  $x_0, y_0$  and  $z_0$ .

Then for  $n = k + 2$ , such integers also exist because

$$(14x_0)^2 + (14y_0)^2 + (14z_0)^2 = 14^{k+2}$$

By the principle of induction, the result follows.

- 20\*.** Let  $a_n = x^n + \frac{1}{x^n}$  then we have

$$a_1 = x + \frac{1}{x}$$

and

$$a_2 = x^2 + \frac{1}{x^2} = \left(x + \frac{1}{x}\right)^2 - 2$$

which are integers.

Suppose that  $a_k$  and  $a_{k+1}$  are integers for some integers  $k \geq 1$ . We will prove that  $a_{k+2}$  is integer.

We can rewrite  $a_{k+2}$  as

$$a_{k+2} = \left(x + \frac{1}{x}\right)a_{k+1} - a_k = ka_{k+1} - a_n$$

We deduce from  $a_k$  and  $a_{k+1}$  are integers that  $a_{k+2}$  is integer.

By use Mathematical induction we have the required claim.

- 22\*.** To show that the quicksort algorithm works, use induction (surprise!) First, we'll show that it works for sets of size zero or of size 1. Those sets are already sorted, so there is nothing to do, and since they fall under the first case of the algorithm which says to do nothing, we are in business.

If the quicksort algorithm works for all sets of numbers of size  $k$  or smaller, then if we start with a list of size  $k + 1$ , since we pick out one element for comparisons and divide the rest of the set into two subsets, it is obvious that each of the subsets has size smaller than or equal to  $k$ . Since the algorithm works on all of those, we know that the full algorithm works since the numbers smaller than the test number are sorted, then comes the test number, then comes a sorted list of all the numbers larger than it.

This algorithm is heavily used in the real world. Surprisingly, the algorithm's performance is worst if the original set is already in order. Can you see why?

- 23\*.** By induction on  $n$ . Base case:  $n = 4$ . 2 two's, done. Induction step: suppose the machine can already handle  $n \geq 4$  dollars. To produce  $n + 1$  dollars, we proceed as follows.

Case 1: The  $n$  dollar output contains a five. Then we can replace the five by 3 two's to get  $n + 1$  dollars.

Case 2: The  $n$  dollar output contains only two's. Since  $n \geq 4$ , there must be at least 2 two's. Remove 2, and replace them by 1 five.

**24\***. Again, this is an easy induction proof. If there is only one disk on a post, you can immediately move it to another post and you are done.

If you know that it is possible to move  $k$  disks to another post, then if you initially have  $k + 1$  disks, move the top  $k$  of them to a different post, and you know that this is possible. Then you can move the largest disk on the bottom to the other empty post, followed by a movement of the  $k$  disks to that other post.

This method, which can be shown to be the fastest possible, requires  $2^k - 1$  steps to move  $k$  disks. This can also be shown by induction — if  $k = 1$ , it requires  $2^1 - 1 = 1$  move. If it's true for stacks of size up to  $k$  disks, then to move  $k + 1$  requires  $2^k - 1$  (to move the top  $k$  to a different post) then 1 (to move the bottom disk), and finally  $2^k - 1$  (to move the  $k$  disks back on top of the moved bottom).

The total for  $k + 1$  disks is thus  $(2^k - 1) + 1 + (2^k - 1) = 2 \cdot 2^k - 1 = 2^{k+1} - 1$ .

The above proof doesn't actually spell out an algorithm to solve the towers of Hanoi problem, but here is such an algorithm. You may be interested in trying to show that the following method always works:

Suppose the posts are numbered 1, 2, and 3, and the disks begin on post 1. Take the smallest disk and move it every other time. In other words, moves 1, 3, 5, 7, et cetera, are all of the top disk. Move the top disk in a cycle — first to post 2, then 3, then 1, then 2, then 3, then 1,... On even moves, make the only possible move that does not involve the smallest disk. This will solve the problem.

**26\***. Obviously, the inequality holds for  $n = 1$ .

Suppose the inequality holds for  $n = 1, 2, \dots, k$  for some positive integer  $k$ .

Then, by adding the inequalities

$$\begin{aligned} a_1 &\geq a_1 \\ a_1 + \frac{a_2}{2} &\geq a_2 \\ &\vdots \\ a_1 + \frac{a_2}{2} + \dots + \frac{a_k}{k} &\geq a_k \end{aligned}$$

we get  $ka_1 + (k-1)\frac{a_2}{2} + \dots + \frac{a_k}{k} \geq a_1 + a_2 + \dots + a_k$

**29.** Starting with 1 gives 4, then 2, then 1, then repeats. 2 is the same story. Starting with 3 gives 10, 5, 16, 8, 4, 2, 1, then repeats. 4 repeats quickly. Starting with 5 gives 16, 8, uh-oh. Starting with 6 gives 3, then we've been here before. Starting with 7 gives 22, 11!

**30.** Prove  $u_n = 3 \cdot 2^{n-1}$  by induction.

**37\***. We have  $\frac{2}{k(k+2)} = \frac{1}{k} - \frac{1}{k+2}$  where  $k = \overline{1, n}$ . Thus

$$2A = \dots = \frac{3}{2} \left( \frac{1}{2n+1} + \frac{1}{2n+2} \right).$$

Therefore  $\frac{4n+3}{(2n+1)(2n+2)} = \frac{199}{9900}$ .  $n = 49$

**39\***. The requirement that  $a_n$  be the smallest positive value of  $n = a_k^2$  for  $k < n$  is equivalent to determining the largest value of  $a_k$  such that  $a_k^2 < n$ . For  $n = 3$ , use either  $a_1 = a_2 = 1$  to find  $a_3 = 3 - 1^2 = 2$ . For  $n = 4$ , the strict inequality eliminates  $a_3$ , so  $a_4 = 4 - 1^2 = 3$ , but  $a_3$  can be used to compute  $a_5 = 5 - 2^2 = 1$ . In fact, until  $n = 10$ , the largest allowable prior value of  $a_k$  is  $a_3 = 2$ , yielding the values  $a_6 = 2, a_7 = 3, a_8 = 4, a_9 = 5$ . In general, this pattern continues: from  $n = m^2 + 1$  until  $n = (m+1)^2$ , the values of  $a_n$  increase from 1 to  $2m+1$ .

Let  $S_m = 1 + 2 + \cdots + (2m+1)$ . Then the problem reduces computing  $S_0 + S_1 + \cdots + S_6 + 1$ , because  $a_{49} = 49 - 6^2$  while  $a_{50} = 50 - 7^2 = 1$  and  $S_m = \frac{(2m+1)(2m+2)}{2} = 2m^2 + 3m + 1$ , so

$$\begin{aligned} S_0 + S_1 + S_2 + S_3 + S_4 + S_5 + S_6 &= 1 + 6 + 15 + 28 + 45 + 66 + 91 \\ &= 252 \end{aligned}$$

Therefore the desired sum is  $252 + 1 = 253$ .

**41\***. We have  $u_2 = 1$ , and  $u_{n+1}^2 - 10u_n u_{n+1} + 25u_n^2 = 24u_n^2 + 1$ , or

$$u_{n+1}^2 - 10u_n u_{n+1} + u_n^2 - 1 = 0$$

Replace  $u_{n+1}$  by  $u_{n-1}$  we have

$$u_{n-1}^2 - 10u_n u_{n-1} + u_n^2 - 1 = 0$$

Therefore,  $u_{n-1}$  and  $u_{n+1}$  are roots of the equation  $t^2 - 10u_n t + u_n^2 - 1 = 0$ . By Viét's theorem, we have  $u_{n-1} + u_{n+1} = 10u_n$  or

$$u_{n+1} = 10u_n - u_{n-1}$$

Note that,  $u_1 = 0$  and  $u_2 = 1$ , thus term of this sequence is integer number.

**42\***. The recurrence  $a_n = 2a_{n-1}^2 - 1$  resembles the double-angle identity  $\cos^2 \theta = 2\cos 2\theta - 1$ . With that in mind, we shall set

$$\theta = \arccos a_0 = \arccos \frac{4}{5}$$

The we have

$$a_1 = 2a_0^2 - 1 = 2\cos^2 \theta - 1 = \cos 2\theta$$

In the same way, we find that

$$a_k = \cos 2^k \theta$$

We're interested in the product

$$\cos \theta \cos 2\theta \cos 4\theta \dots \cos 2^{n-1} \theta.$$

Using the double-angle identity  $\sin 2\theta = 2 \sin \theta \cos \theta$ , we get the extended identity

$$\sin \theta \cos \theta \cos 2\theta \cos 4\theta \dots \cos 2^{n-1} \theta = \sin 2^n \theta$$

Because  $a_k = \cos 2^k \theta$ , we have

$$2^n (\sin \theta) a_0 a_1 \dots a_{n-1} = \sin 2^n \theta$$

Because  $\cos \theta = \frac{4}{5}$ , we have  $\sin \theta = \frac{3}{5}$ . Therefore

$$a_0 a_1 a_2 \dots a_{n-1} = \frac{5 \sin 2^n \theta}{3 \cdot 2^n}$$

Because  $\sin 2^n \theta \leq 1$ , we have

$$a_0 a_1 a_2 \dots a_{n-1} \leq \frac{5}{3 \cdot 2^n}$$

Hence  $c \leq \frac{5}{3}$ . We can check by computer that

$$\sin 2^{36} \theta > 0.999.$$

Then the analysis above shows that

$$c \geq \frac{5}{3} \sin 2^{36} \theta > \frac{5}{3} \cdot 0.999 = 1.665.$$

So  $100c$  is greater than 166.5 and at most  $\frac{500}{3} = 166\frac{2}{3}$ . Hence  $100c$  rounded to the nearest integer is 167.

**54.** Let  $d$  be the common difference and three terms are  $x - d, x, x + d$ . We have

$$\begin{cases} x - d + x + x + d = 9 \\ (x - d)^2 + x^2 + (x + d)^2 = 125 \end{cases}$$

Therefore, we have  $x = 3, d = \pm 7$ .

**55.** Let  $x - 3a, x - a, x + a, x + 3a$  be side lengths of a quadrilateral produce. Since the perimeter is 60 cm, we have:

$$(x - 3a) + (x - a) + (x + a) + (x + 3a) = 60 \Leftrightarrow x = 15$$

The longest side has a length of 24 cm, thus  $x + 3a = 24 \Leftrightarrow a = 3$ . Hence, the other side lengths are 6 cm, 12 cm and 18 cm.

**57.** Assume that these roots  $x_1, x_2, x_3$  form an arithmetic sequence. We can set:  $x_1 = x_0 - d, x_2 = x_0, x_3 = x_0 + d$  ( $d \neq 0$ ).

Then

$$f(x) = (x - x_1)(x - x_2)(x - x_3) = x^3 - 3x_0x^2 + (3x_0^2 - d^2)x - x_0^3 + x_0d^2$$

Therefore,  $x_0 = -1, m = n$ .

58. Setting  $t = x^2$  we have

$$t^2 - (3m+5)t + (m+1)^2 = 0$$

If this equation has two roots  $0 < t_1 < t_2$  then four roots of given equation are

$$-\sqrt{t_2}, -\sqrt{t_1}, \sqrt{t_1}, \sqrt{t_2}$$

Therefore  $t_2 = 9t_1$  and  $m = 5$  or  $m = -\frac{25}{19}$ .

60. We have,  $\frac{1}{b+c}, \frac{1}{c+a}, \frac{1}{a+b}$  is an arithmetic sequence if and only if

$$\begin{aligned} \frac{1}{c+a} - \frac{1}{b+c} &= \frac{1}{a+b} - \frac{1}{c+a} \\ \Leftrightarrow \frac{b-a}{(c+a)(b+c)} &= \frac{c-b}{(a+b)(c+a)} \\ \Leftrightarrow b^2 - a^2 &= c^2 - b^2 \\ \Leftrightarrow 2b^2 &= a^2 + c^2 \end{aligned}$$

Obviously,  $a^2, b^2, c^2$  is an arithmetic sequence if and only if  $2b^2 = a^2 + c^2$ . Hence, we have the required claim.

64\*. Suppose that  $n^2$  is a perfect square in an infinite arithmetic progression  $A$  where common difference  $r > 0$ . Then

$$(n+r)^2 = n^2 + (2n+r)r$$

is a larger perfect square in  $A$ . So for each perfect square in  $A$ , there is a larger perfect square in  $A$ ; applying this statement to the larger perfect square gives us a still larger perfect square in  $A$ . Continuing in this manner, we see that there are infinitely many perfect squares in  $A$ .

65\*. Assume by contradiction that there exists positive integers

$$a_1 < a_2 < \dots < a_n < a_{n+2} < \dots$$

such that

$$a_1^2 < a_2^2 < \dots < a_n^2 < a_{n+2}^2 < \dots$$

is an arithmetic progression with common difference  $r$ :

$$r = a_2^2 - a_1^2 = a_3^2 - a_2^2 = \dots = a_n^2 - a_{n-1}^2$$

It follows that

$$(a_n - a_{n-1})(a_n + a_{n-1}) = (a_{n+1} - a_n)(a_{n+1} + a_n), \quad n = 2, 3, 4, \dots$$

Since  $a_{n-1} < a_n < a_{n+1}$  we have  $a_{n+1} + a_n > a_n + a_{n-1}$  so the above equality yields

$$a_2 - a_1 > a_3 - a_2 > a_4 - a_3 > \dots > a_n - a_{n-1} > \dots > 0$$

which is clearly impossible.

**66\***. We have

$$\begin{aligned} S &= (1-2)(1+2) + (3-4)(3+4) + \cdots + (2n-1-2n)(2n-1+2n) \\ &= -3 - 7 - \cdots - (4n-1) \\ &= (-3) + (-7) + \cdots + (-4n+1) \end{aligned}$$

This is the sum of  $n$  first terms of an arithmetic sequence where the first term  $u_1 = -3$  and common difference  $d = -4$ . Hence,  $S = \frac{n(u_1+u_n)}{2} = \frac{n(-3+(-4n+1))}{2} = -2n-1$ .

**68.** Since  $\frac{a_{n+1}}{a_n} = \frac{2 \cdot 3^{n+1}}{2 \cdot 3^n} = 3 > 1$  we have  $(a_n)$  is an geometric sequence where the common ratio is  $q = 3$ .

**69.**  $(a-c)^2 + (b-c)^2 + (b-d)^2 - (a-d)^2 = 0$ .

**70.** The ratios of successive terms will be equal since we have a geometric sequence. So

$$\frac{a}{b} = \frac{b}{a+b} = \frac{a+b}{ab} \quad (*)$$

Therefore,

$$a^b + ab = b^2 \Leftrightarrow \left(\frac{b}{a}\right)^2 - \frac{b}{a} - 1 = 0 \Leftrightarrow \frac{b}{a} = \frac{1+\sqrt{5}}{2}$$

where we have chosen the positive root since  $a$  and  $b$  are positive. Also from  $(*)$ ,

$$\frac{a}{b} = \frac{a+b}{ab} \Leftrightarrow a^2 = a+b \Leftrightarrow a = 1 + \frac{b}{a} = \frac{3+\sqrt{5}}{2}.$$

**71.** Có  $u_1 + u_2 = u_1 + \frac{1}{4}u_1 = 24 \Leftrightarrow u_1 + \frac{1}{4}u_1^2 - 24 = 0 \Leftrightarrow u_1 = -12 \vee u_1 = 8$ .

**91\***. Suppose that the lengths of the sides of it is  $0 < a \leq b \leq c$ , which forms a geometric sequence. Thus  $b^2 = ac$ . Using cosin rule we have

$$b^2 = a^2 + c^2 - 2ac \cos B \Rightarrow ac = a^2 + c^2 - 2ac \cos B \Leftrightarrow \cos B = \frac{a^2 + c^2}{2ac} - \frac{1}{2}$$

Beside that,  $a^2 + c^2 \geq 2ac$ . Thus  $\cos B \geq 1 - \frac{1}{2} = \frac{1}{2}$  or  $B \leq 60^\circ$ . Notice that  $a \leq b$  then  $A \leq 60^\circ$ . Hence we have the required claim.

**92.** We have  $\begin{cases} xy = 3^2 \\ x^4 = y\sqrt{3} \end{cases} \Leftrightarrow \begin{cases} x = \sqrt{3} \\ y = 3\sqrt{3} \end{cases}$

**95\***. Denote the geometric sequence is  $(v_n)$  and the arithmetic sequence is  $(u_n)$  then we have

$$\begin{cases} v_1 + v_2 + v_3 = 93 \\ v_1 = u_1 \\ u_1 + d = v_1 q \\ u_1 + 6d = v_1 q^2 \end{cases} \Leftrightarrow \begin{cases} v_1(1+q+q^2) = 93 \\ d = u_1(q-1) \\ 6d = u_7 - u_1 = u_1(q^2-1) \end{cases} \Leftrightarrow \begin{cases} u_1(1+q+q^2) = 93 \\ d = u_1(q-1) \\ u_1(q-1) = \frac{1}{6}u_1(q^2-1) \end{cases}$$

Therefore, three required numbers are 3, 15, 75.

**97\***. We have

$$\begin{aligned}
 S &= \left(4 + 2 + \frac{1}{4}\right) + \left(16 + 2 + \frac{1}{16}\right) + \dots + \left(2^{2n} + 2 + \frac{1}{2^{2n}}\right) \\
 &= (4 + 16 + \dots + 2^{2n}) + 2n + \left(\frac{1}{4} + \frac{1}{16} + \dots + \frac{1}{2^{2n}}\right) \\
 &= 4 \cdot \frac{4^{n-1} - 1}{3} + 2n + \frac{1}{4} \cdot \frac{2^{\frac{1}{2n}} - 1}{\frac{1}{4} - 1} \\
 &= 4 \cdot \frac{4^n - 1}{3} + 2n + \frac{1}{3} \cdot \frac{2^{2n} - 1}{2^{2n}} = 2n + \frac{4^n - 1}{3} \cdot \frac{4 \cdot 4^n + 1}{4^n} = 2n + \frac{(4^n - 1)(4^{n+1} + 1)}{3 \cdot 4^n}
 \end{aligned}$$

**98\***. We have

$$\begin{aligned}
 S &= 3(1 + 11 + 111 + \dots + 11\dots1) = 3\left(\frac{10-1}{9} + \frac{10^2-1}{9} + \dots + \frac{10^n-1}{9}\right) \\
 &= \frac{3}{9}(10 + 10^2 + \dots + 10^n - n) = \frac{1}{3}\left(10 \cdot \frac{10^n-1}{10-1} - n\right) \\
 &= \frac{1}{27}(10^{n+1} - 10 - 9n)
 \end{aligned}$$

**106.** The common ratio in the geometric sequence  $1, x, y$  is  $\frac{y}{x} = x$ , so  $y = x^2$ . The arithmetic sequence  $x, y, 3$  has a common difference, so  $y - x = 3 - y$ . Substituting  $y = x^2$  in the equation yields

$$\begin{aligned}
 x^2 - x &= 3 - x^2 \\
 2x^2 - x - 3 &= 0,
 \end{aligned}$$

from which  $x = \frac{3}{2}$  or  $-1$ . The respective values of  $y$  are  $y = x^2 = \frac{9}{4}$  or  $1$ . Thus the possible values of  $x + y$  are  $\frac{15}{4}$  and  $0$ , so the answer is  $\frac{15}{4}$ .

**107.**  $1, \frac{2}{3}, \frac{1}{3}$  và  $\frac{1}{3}, \frac{2}{3}, 1$ .

**108.** Since  $a, b, c$  form an arithmetic sequence, then we can write  $a = b - d$  and  $c = b + d$  for some real number  $d$ .

Since  $a + b + c = 60$ , then  $(b - d) + b + (b + d) = 60$  or  $3b = 60$  or  $b = 20$ .

Therefore, we can write  $a, b, c$  as  $20 - d, 20, 20 + d$ . (We could have written  $a, b, c$  instead as  $a, a + d, a + 2d$  and arrived at the same result.)

Thus,  $a - 2 = 20 - d - 2 = 18 - d$  and  $c + 3 = 20 + d + 3 = 23 + d$ , so we can write  $a - 2, b, c + 3$  as  $18 - d, 20, 23 + d$ . Since these three numbers form a geometric sequence, then

$$\begin{aligned}
 \frac{20}{18-d} &= \frac{23+d}{20} \\
 20^2 &= (23+d)((18-d)) \\
 d^2 + 5d - 14 &= 0
 \end{aligned}$$

Therefore,  $d = -7$  or  $d = 2$ .

If  $d = -7$ , then  $a = 27, b = 20$  and  $c = 13$ .

If  $d = 2$ , then  $a = 18, b = 20$  and  $c = 22$ . (We can check that, in each case,  $a - 2, b, c + 3$  is a geometric sequence.)

**109.** Since  $x + 6y, 5x + 2y, 8x + y$  form an arithmetic sequence we have  $x = 3y$ . Since  $x + \frac{5}{3}, y - 1, 2x - 3y$  form a geometric sequence we have  $x = -3, y = -1$  and  $x = \frac{3}{8}, y = \frac{1}{8}$ .

**115.** We have

$$\begin{cases} a^2 = b^2 + c^2 \\ \frac{2}{3}b^2 = ac \end{cases} \Leftrightarrow \begin{cases} a^2 = b^2 + c^2 \\ b^2 = \frac{3}{2}ac \end{cases}$$

So  $a^2 = \frac{3}{2}ac + c^2 \Leftrightarrow 2a^2 = 3ac + 2c^2 \Leftrightarrow (2a + c)(a - 2c) = 0 \Rightarrow a = 2c$ . Beside that  $\cos B = \frac{c}{a} = \frac{1}{2}$  hence  $B = 60^\circ, C = 30^\circ$ .

**120\*.** Notice that  $\frac{2k+1}{2k+2} < \frac{\sqrt{2k+1}}{\sqrt{2k+3}}$ .

**121\*.** Using  $a^{k+1} + b^{k+1} - (a^k b + ab^k) = a^k(a - b) - b^k(a - b) = (a - b)(a^k - b^k) \geq 0$

**124.** Using  $|a + b| \leq |a| + |b|$ .

**125\*.** Let  $a_2 - a_1 = d$  and  $\frac{b_2}{b_1} = r$ . Using  $a = a_1$  and  $b = b_1$ , write the system of equation

$$\begin{cases} a + b & = 1 \\ (a + d) + br & = 4 \\ (a + 2d) + br^2 & = 15 \\ (a + 3d) + br^3 & = 2 \end{cases}$$

Subtract the first equation from the second, the second from the third, and the third from the fourth to obtain three equations:

$$\begin{cases} d + b(r - 1) & = 3 \\ d + b(r^2 - r) & = 11 \\ d + b(r^3 - r^2) & = -13 \end{cases}$$

Notice that the  $a$  terms have canceled. Repeat to find the second differences:

$$\begin{cases} b(r^2 - 2r + 1) & = 8 \\ b(r^3 - 2r^2 + r) & = -24 \end{cases}$$

Now divide the second equation by the first to obtain  $r = -3$ . Substituting back into either of these two last equations yields  $b = \frac{1}{2}$ . Continuing in the same vein yields  $d = 5$  and  $a = \frac{1}{2}$ . Then  $a_5 = \frac{41}{2}$  and  $b_5 = \frac{81}{2}$ , so  $c_5 = 61$ .

**126\***. We consider choosing the three numbers all at once. We list the possible sets of three numbers that can be chosen:

$$\begin{array}{ccccccccc} \{1,2,3\} & \{1,2,4\} & \{1,2,5\} & \{1,3,4\} & \{1,3,5\} \\ \{1,4,5\} & \{2,3,4\} & \{2,3,5\} & \{2,4,5\} & \{3,4,5\} \end{array}$$

We have listed each in increasing order because once the numbers are chosen, we arrange them in increasing order. There are 10 sets of three numbers that can be chosen. Of these 10, the 4 sequences 1,2,3 and 1,3,5 and 2,3,4 and 3,4,5 are arithmetic sequences. Therefore, the probability that the resulting sequence is an arithmetic sequence is  $\frac{2}{5}$ .

## Chương 4

# Kết luận

Qua báo cáo này, chúng tôi mong muốn giải quyết được những vấn đề sau:

- Giúp giáo viên có một tài liệu tham khảo đầy đủ, chính xác; đặc biệt là có một hệ thống các bài toán với lời giải mẫu, cách trình bày chuẩn về nội dung dãy số, cấp số và phương pháp quy nạp.
- Giúp học sinh có một tài liệu tham khảo về các trình bày lời giải Toán bằng tiếng Anh; có hệ thống bài tập tham khảo phong phú từ mức độ cơ bản đến rất khó phục vụ cho các kì thi học sinh giỏi.
- Tiết kiệm thời gian, công sức trong việc tiếp cận với kiến thức về dãy số, phương pháp quy nạp trong chương trình Toán của các nước sử dụng tiếng Anh.
- Là một tài liệu tham khảo cho việc luyện thi SAT, ACT...

Việc giảng dạy Toán sao cho dễ hiểu, hấp dẫn đã khó; việc giảng dạy Toán bằng tiếng Anh còn khó hơn. Tuy nhiên, bên cạnh những khó khăn thì việc giảng dạy Toán bằng tiếng Anh cũng có những lợi thế nhất định, đúng như câu nói "Toán học là ngôn ngữ của vũ trụ", học sinh nắm chắc kiến thức Toán có thể dựa vào các công thức trong đề bài để đoán được nghĩa các từ vựng, mà không cần một kiến thức tiếng Anh quá sâu sắc. Như vậy, bản thân việc học Toán bằng tiếng Anh đã mang lại ba lợi ích:

- Củng cố và đào sâu các kiến thức Toán liên quan.
- Hỗ trợ cho việc học tiếng Anh của học sinh, đặc biệt là phần giao tiếp.
- Thấy được lợi ích của việc học tiếng Anh, giúp học sinh tiếp cận với kho tri thức khổng lồ của nhân loại trên mạng internet.

Bản thân chúng tôi, trong một nỗ lực để đưa việc giảng dạy Toán bằng tiếng Anh đến gần với thực tế hơn, đã giao toàn bộ bài tập phân môn "Đại số và Giải tích 11" bằng tiếng Anh cho học

sinh. Khi đó, học sinh có hai nhiệm vụ: dịch đề bài sang tiếng Việt và giải quyết bài toán; bên cạnh đó, chúng tôi cũng khuyến khích học sinh trình bày lời giải bằng tiếng Anh.

Hì vọng tài liệu này giúp các bạn đồng nghiệp phần nào trong việc giảng dạy Toán bằng tiếng Anh. Được biên soạn trong một thời gian gấp rút nên tài liệu này không tránh khỏi sai sót, rất mong nhận được góp ý của các bạn đồng nghiệp và các em học sinh.